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(54) **SYSTEMS AND METHODS FOR THE RECYCLING OF LITHIUM FROM BATTERY WASTE**

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(57) **ABSTRACT**

Embodiments described herein relate to recycling of spent lithium battery material. In some aspects, a method can include suspending a lithium source in a solvent containing an oxidation reagent to extract lithium, forming an extracted lithium solution, separating the extracted lithium solution from residual solids of a lithium source, purifying the extracted lithium solution by precipitating and filtering impurities, and precipitating the lithium in the purified lithium solution to generate lithium carbonate (Li_2CO_3). In some embodiments, the method can further include preprocessing the lithium source to improve kinetics of the lithium extraction. In some embodiments, the preprocessing can include a cutting or shredding step to downsize the lithium source. In some embodiments, the lithium source can include lithium-ion battery waste. In some embodiments, the oxidation reagent can include sodium persulfate ($Na_2S_2O_8$), potassium persulfate ($K_2S_2O_8$), ammonium persulfate $(NH_4)_2S_2O_8$, hydrogen peroxide (H_2O_2), ozone (O_3), and/or nitrous oxide (N_2O).

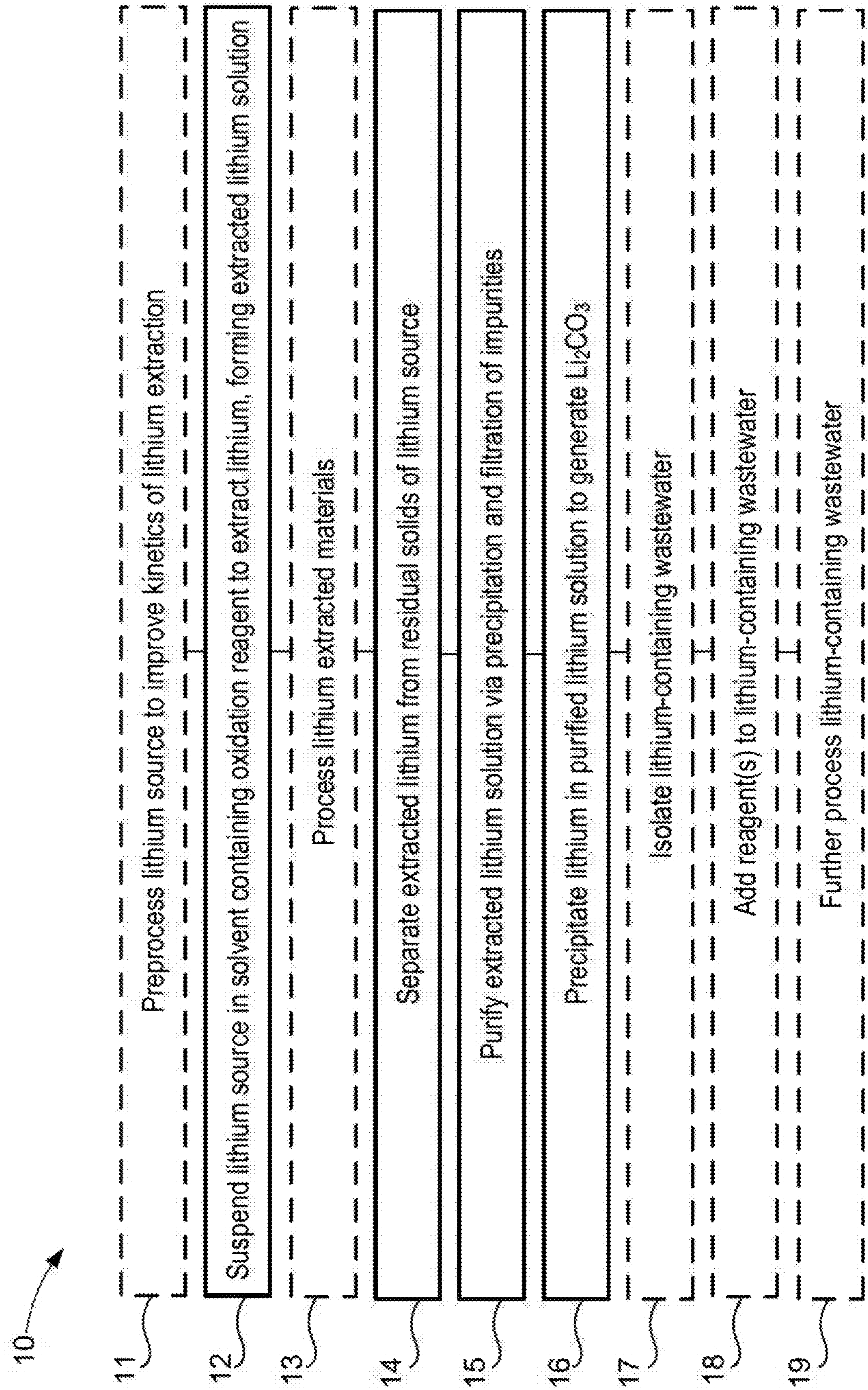
10
11 Preprocess lithium source to improve kinetics of lithium extraction
12 Suspend lithium source in solvent containing oxidation reagent to extract lithium, forming extracted lithium solution
13 Process lithium extracted materials
14 Separate extracted lithium from residual solids of lithium source
15 Purify extracted lithium solution via precipitation and filtration of impurities
16 Precipitate lithium in purified lithium solution to generate Li_2CO_3
17 Isolate lithium-containing wastewater
18 Add reagent(s) to lithium-containing wastewater
19 Further process lithium-containing wastewater

FIG. 1

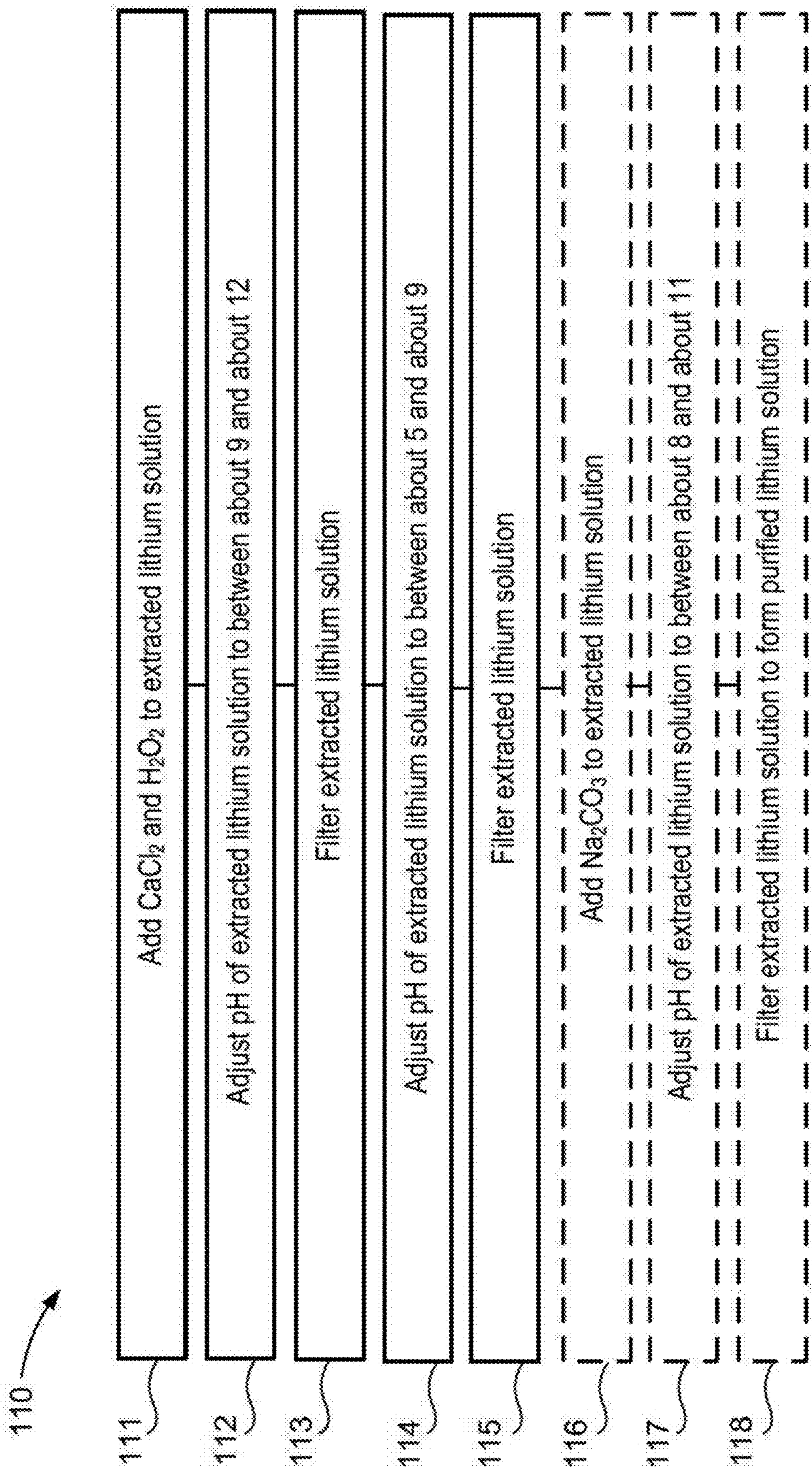
110
111
Add CaCl2 and H2O2 to extracted lithium solution
112
Adjust pH of extracted lithium solution to between about 9 and about 12
113
Filter extracted lithium solution
114
Adjust pH of extracted lithium solution to between about 5 and about 9
115
Filter extracted lithium solution
116
Add Na2CO3 to extracted lithium solution
117
Adjust pH of extracted lithium solution to between about 8 and about 11
118
Filter extracted lithium solution to form purified lithium solution

FIG. 2

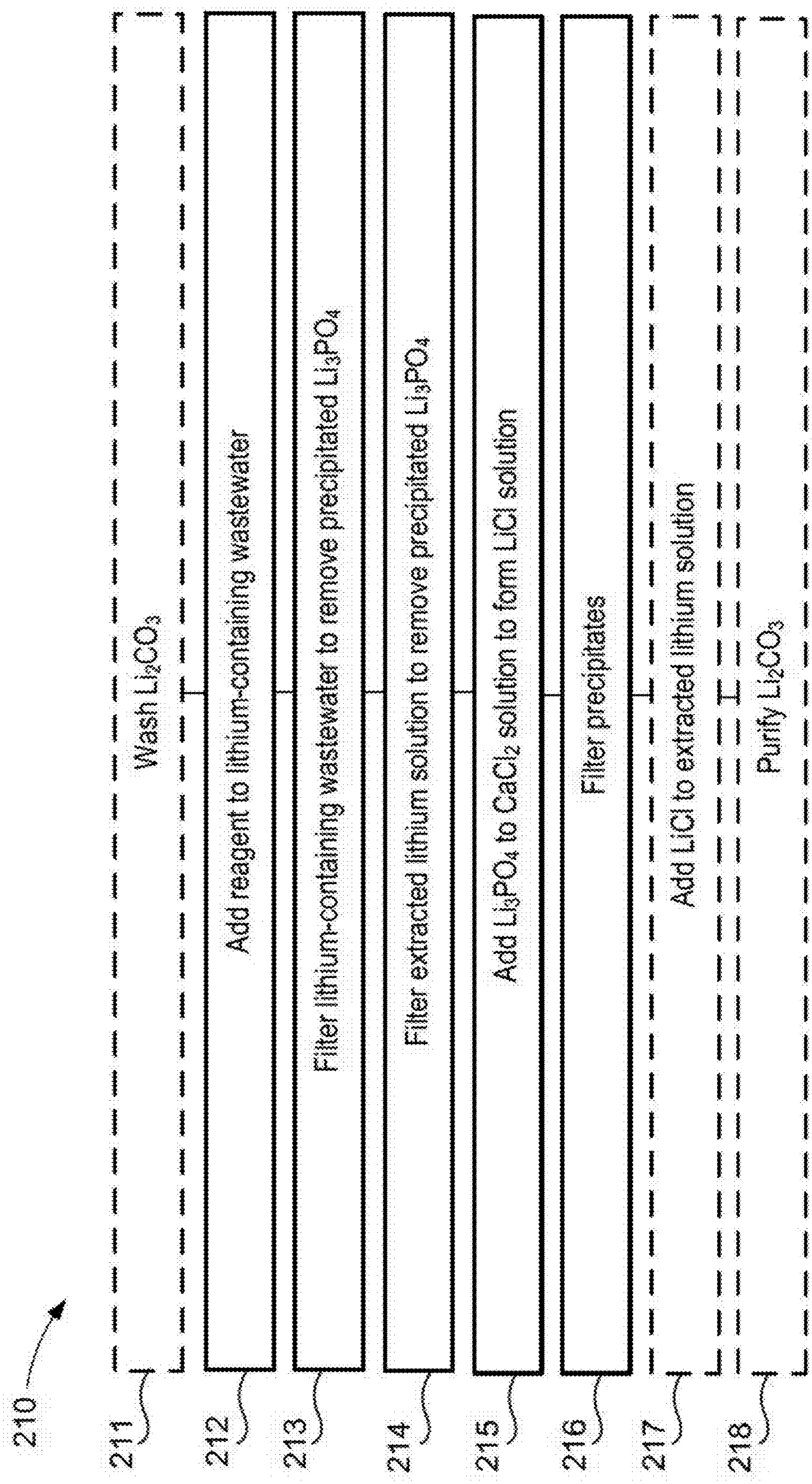
210
211
Wash Li_2CO_3
212
Add reagent to lithium-containing wastewater
213
Filter lithium-containing wastewater to remove precipitated Li_3PO_4
214
Filter extracted lithium solution to remove precipitated Li_3PO_4
215
Add Li_3PO_4 to $CaCl_2$ solution to form LiCl solution
216
Filter precipitates
217
Add LiCl to extracted lithium solution
218
Purify Li_2CO_3

FIG. 3

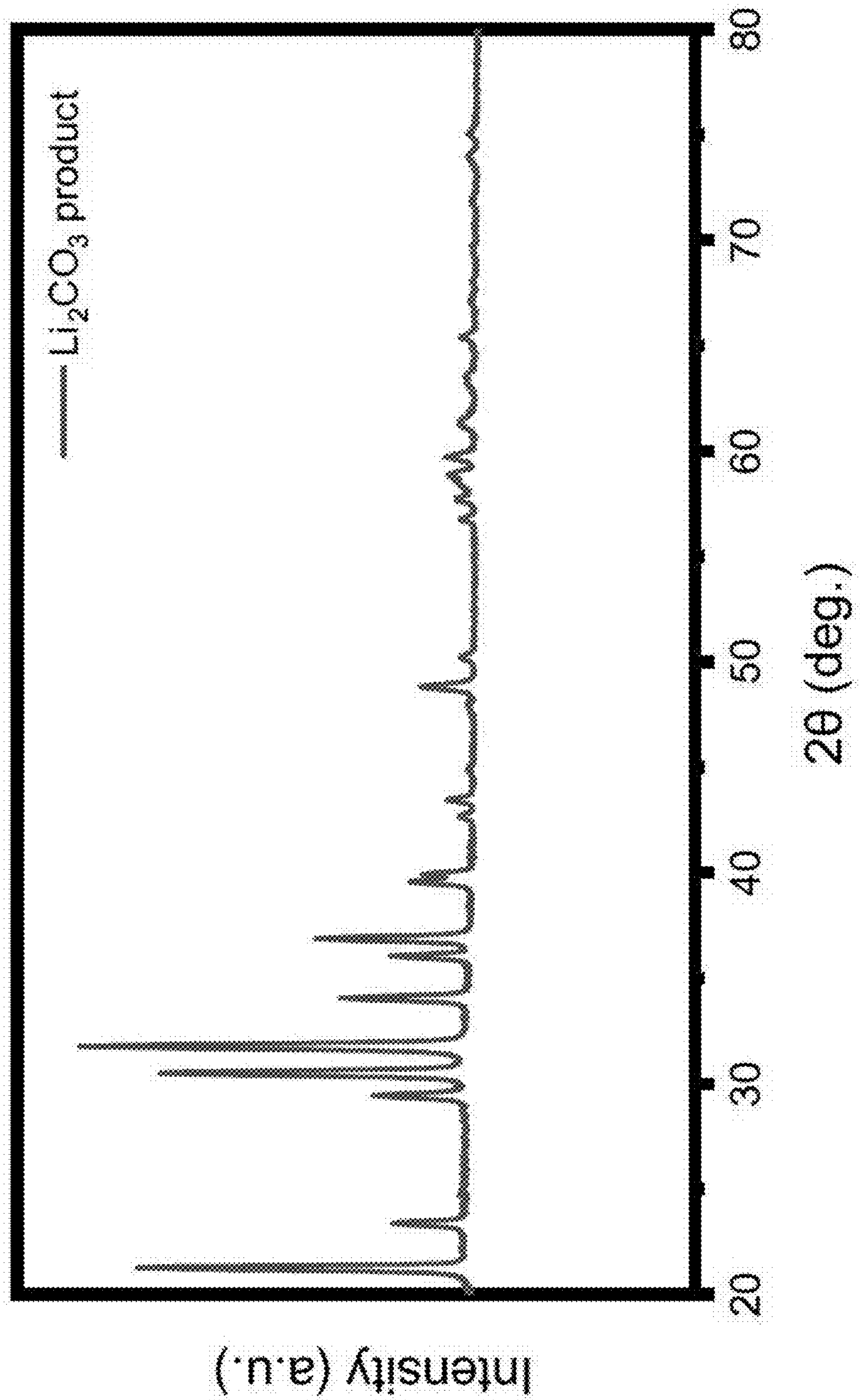

FIG. 4

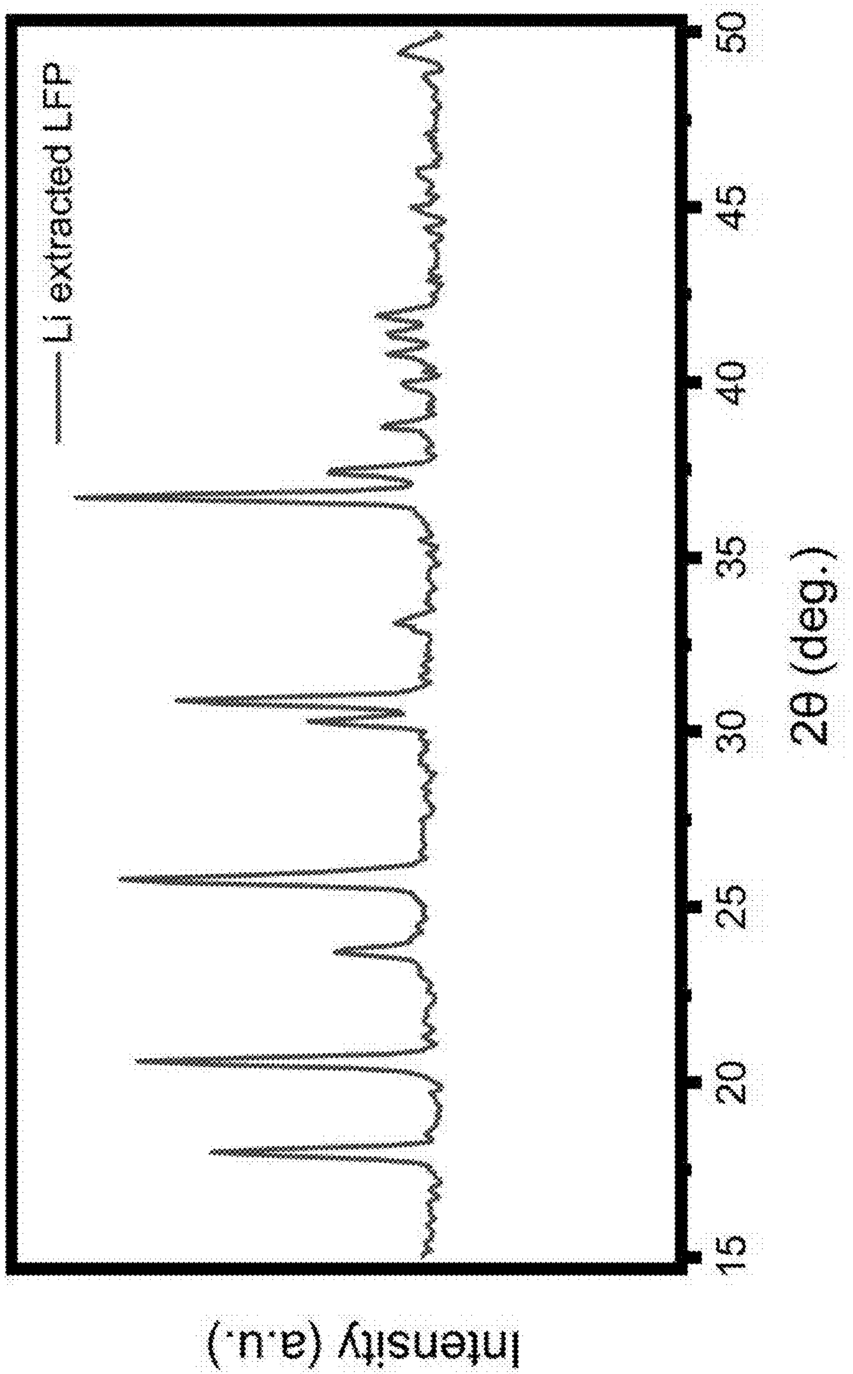

FIG. 5

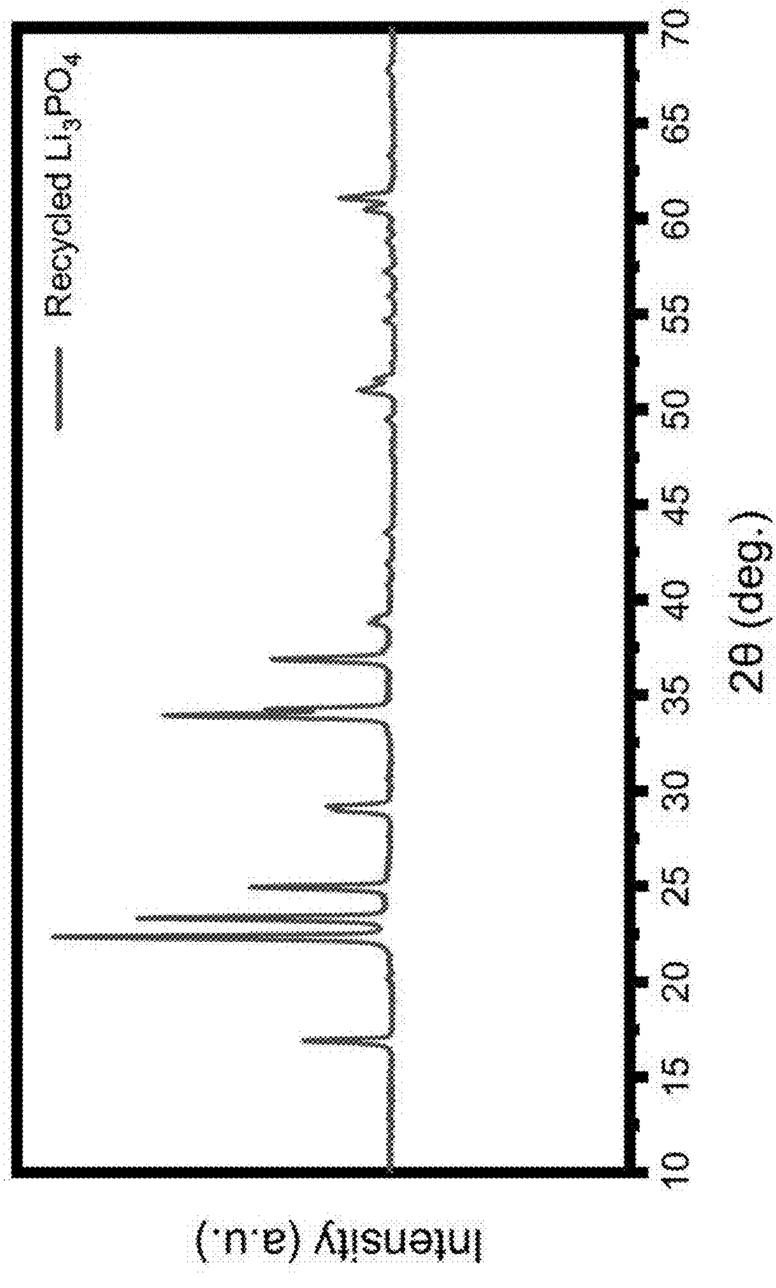
Recycled Li_3PO_4
Intensity (a.u.)
2θ (deg.)
10 15 20 25 30 35 40 45 50 55 60 65 70

FIG. 6

SYSTEMS AND METHODS FOR THE RECYCLING OF LITHIUM FROM BATTERY WASTE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. application Ser. No. 18/889,049, entitled "Systems and Methods for the Recycling of Lithium from Battery Waste," filed Sep. 18, 2024, which is a continuation of U.S. application Ser. No. 18/592,728, entitled "Systems and Methods for the Recycling of Lithium from Battery Waste," filed Mar. 1, 2024, now U.S. Pat. No. 12,129,180, which claims priority to and the benefit of U.S. Provisional Application No. 63/488,378, entitled "Systems and Methods for the Recycling of Lithium from Battery Waste," filed Mar. 3, 2023, and U.S. Provisional Application No. 63/469,950, entitled "Systems and Methods for the Recycling of Lithium from Battery Waste," filed May 31, 2023, the disclosures of which are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

[0002] Embodiments described herein relate systems and methods of recycling lithium from spent battery material.

BACKGROUND

[0003] Lithium-ion batteries (LIBs) are widely implemented for portable electronic devices, electric vehicles, and grid energy storage due to their low self-discharge rate, high energy and power density, and long cycle life. The market for LIBs will continue to grow in the future. Among the cathode active materials, olivine-type lithium iron phosphate ($LiFePO_4$, also referred to as LFP), and its derivatives, have attracted much attention and have several distinct advantages over their counterparts. LFP and its derivatives, when used as cathode materials in LIBs, are intrinsically safer, cheaper, and highly durable when compared to other cathode materials. However, the LIB production process results in the formation of undesired by-products. Capture and use of these by-products can greatly improve the efficiency of the overall LIB production process.

SUMMARY

[0004] Embodiments described herein relate to recycling of spent lithium battery material. In some aspects, a method can include suspending a lithium source in a solvent containing an oxidation reagent to extract lithium, forming an extracted lithium solution, separating the extracted lithium solution from residual solids of a lithium source, purifying the extracted lithium solution by precipitating and filtering impurities, and precipitating the lithium in the purified lithium solution to generate lithium carbonate (Li_2CO_3). In some embodiments, the method can further include preprocessing the lithium source to improve kinetics of the lithium extraction. In some embodiments, the preprocessing can include a cutting or shredding step to downsize the lithium source. In some embodiments, the lithium source can include lithium-ion battery waste. In some embodiments, the oxidation reagent can include sodium persulfate ($Na_2S_2O_8$), potassium persulfate ($K_2S_2O_8$), ammonium persulfate $(NH_4)_2S_2O_8$, hydrogen peroxide (H_2O_2), ozone (O_3), and/or nitrous oxide (N_2O).

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. **1** is a flow diagram of a method of producing Li_2CO_3 from a lithium source, according to an embodiment.

[0006] FIG. **2** is a flow diagram of a method of processing an extracted lithium solution, according to an embodiment.

[0007] FIG. **3** is a flow diagram of a method of processing lithium-containing wastewater, according to an embodiment.

[0008] FIG. **4** shows the X-ray diffraction (XRD) pattern of lithium carbonate generated from an embodiment compared to a standard lithium carbonate XRD pattern.

[0009] FIG. **5** shows the XRD pattern of Li-extracted iron phosphate generated from an embodiment compared to a standard iron phosphate XRD pattern.

[0010] FIG. **6** shows the XRD pattern of lithium phosphate generated from an embodiment compared to a standard lithium phosphate XRD pattern.

DETAILED DESCRIPTION

[0011] Embodiments described herein relate to systems and methods of recycling spent batteries. Abundant sources of lithium are important for continued generation of new battery materials, including cathode compounds. The demand for lithium has increased globally in recent decades. Industries driving this increase include the battery materials industry, the ceramics industry, and the chemical additives industry. High-purity Li_2CO_3 is the most widely used lithium precursor for the production of different lithium-ion battery cathode materials. Lithium hydroxide (LiOH) is also a common lithium precursor. Extracting lithium as Li_2CO_3 or LiOH is an efficient and economic pathway for the recycling of low cost LFP, and its derivatives, which can include end-of-life, used, scrap, defect LFP batteries, constituent materials, or work-in-process materials.

[0012] Embodiments described herein can develop a high purity Li_2CO_3 product. Methods of producing a high purity Li_2CO_3 process include suspending a lithium source and extracting lithium from the suspended lithium source. Precipitation and filtration can be used to improve the purity of the captured Li_2CO_3. In some embodiments, a lithium source material can originate from an electrode (also referred to as a lithium source electrode), electrode materials, and/or scrap electrode materials. In some embodiments, the lithium source can include at least one chemical composition and/or a chemical structure including lithium. In some embodiments, the lithium source can be included in a mixture or attached to other materials that do not include lithium. In some embodiments, the lithium source can include a component in a system (e.g., a full battery cell, a half battery cell, battery production scrap, battery production work-in-process material). In some embodiments, the component can be shredded, cut, milled, or ground into powder or flakes (e.g., black mass).

[0013] As used in this specification, the singular forms "a," "an," and "the" include plural referents unless the context clearly dictates otherwise. Thus, for example, the term "a member" is intended to mean a single member or a combination of members, "a material" is intended to mean one or more materials, or a combination thereof.

[0014] The term "substantially" when used in connection with "cylindrical," "linear," and/or other geometric relationships is intended to convey that the structure so defined is nominally cylindrical, linear or the like. As one example, a portion of a support member that is described as being "substantially linear" is intended to convey that, although linearity of the portion is desirable, some non-linearity can occur in a "substantially linear" portion. Such non-linearity can result from manufacturing tolerances, or other practical considerations (such as, for example, the pressure or force applied to the support member). Thus, a geometric construction modified by the term "substantially" includes such geometric properties within a tolerance of plus or minus 5% of the stated geometric construction. For example, a "substantially linear" portion is a portion that defines an axis or center line that is within plus or minus 5% of being linear.

[0015] As used herein, the term "set" and "plurality" can refer to multiple features or a singular feature with multiple parts. For example, when referring to a set of electrodes, the set of electrodes can be considered as one electrode with multiple portions, or the set of electrodes can be considered as multiple, distinct electrodes. Additionally, for example, when referring to a plurality of electrochemical cells, the plurality of electrochemical cells can be considered as multiple, distinct electrochemical cells or as one electrochemical cell with multiple portions. Thus, a set of portions or a plurality of portions may include multiple portions that are either continuous or discontinuous from each other. A plurality of particles or a plurality of materials can also be fabricated from multiple items that are produced separately and are later joined together (e.g., via mixing, an adhesive, or any suitable method).

[0016] FIG. **1** is a flow diagram of a method **10** of producing Li_2CO_3, according to an embodiment. Optional steps are shown with dotted boxes. As shown, the method **10** optionally includes preprocessing a lithium source to improve the kinetics of lithium extraction at step **11**. The method **10** further includes suspending the lithium source in a solvent containing an oxidation reagent to extract lithium and form an extracted lithium solution at step **12**. The method **10** optionally includes processing lithium-extracted materials at step **13**. The method **10** further includes separating the extracted lithium from residual solids of the lithium source at step **14**, purifying the extracted lithium solution via precipitation and filtration of impurities at step **15**, and precipitating the lithium in the purified lithium solution to generate Li_2CO_3 at step **16**. The method **10** optionally includes isolating lithium-containing wastewater at step **17**, adding one or more reagents to the lithium-containing wastewater at step **18**, and further processing the lithium-containing wastewater at step **19**.

[0017] Step **11** is optional and includes pre-processing a lithium source to improve the kinetics of lithium extraction. In some embodiments, the preprocessing can include mechanical, chemical, and/or thermal treatment of the lithium source. The preprocessing step can enable or improve oxidation extraction of the lithium source. Higher reaction kinetics lead to a higher oxidation reaction rate and therefore a faster lithium production rate. In some embodiments, the preprocessing can alter the properties of the lithium source to enhance facile lithium extraction. In some embodiments, the pre-processing of the lithium source can include shredding, crushing, and/or pulverizing the lithium source. In some embodiments, the lithium source materials (e.g., batteries) go through a cutting/shredding step to expose the internal cathode materials for the oxidation and extraction at step **12**. In some embodiments, the lithium source materials can be downsized into flakes or pieces via methods such as shredding, cutting, milling, or crushing. In some embodiments, the lithium source materials can go through a comminution process to be downsized to the microscale/nanoscale. In some embodiments, the comminution process can include tamp milling, tooth milling, knife milling, ball milling, jet milling, mortar milling, and crushing.

[0018] In some embodiments, the preprocessing can improve the wetting of the lithium source materials before the oxidation extraction at step **12**. In some embodiments, the lithium source materials can be vacuum-filled to enhance the wetting. In some embodiments, the lithium source materials can be ultrasonicated in water before and/or during the oxidation extraction at step **12**. In some embodiments, the lithium source materials can be heated to a temperature of no more than about 100° C., no more than about 90° C., no more than about 80° C., no more than about 70° C., no more than about 60° C., no more than about 50° C., no more than about 40° C. in water before and/or during the oxidation extraction at step **12**. In some embodiments, the lithium source materials can be heated to a temperature of no more than about 800° C., no more than about 700° C., no more than about 600° C., no more than about 500° C., no more than about 400° C., no more than about 300° C., no more than about 200° C., or no more than about 100° C. in air or another gas mixture (e.g., N_2, Ar) during the preprocessing at step **11**. This heating can serve to remove certain components from the lithium source, such as binder or organics, that can allow for a more facile oxidation process at step **12**.

[0019] Step **12** includes suspending a lithium source in a solvent containing an oxidation reagent to extract lithium, thereby forming an extracted lithium solution. In some embodiments, the lithium source can include a lithium source electrode (i.e., an electrode containing lithium). In some embodiments, the lithium source can originate from a spent battery. In some embodiments, the lithium source electrode can include $LiFePO_4$. In some embodiments, the lithium source electrode can include $Li_xM_yPO_4$, where M is at least one transition metal and x and y are integers. In some embodiments, the lithium source electrode can include the doped derivative of lithium iron phosphate (e.g., $LiM_xFe_{1-x}PO_4$, or $Li_{1-x}M_xPO_4$, wherein M can one or more transition metals and x and y are integers). In some embodiments, the lithium source electrode can include lithium cobalt oxide (LCO), lithium nickel cobalt manganese oxide (NCM or NMC), lithium nickel cobalt aluminum oxide (NCA), and/or lithium manganese oxide (LMO). In some embodiments, the lithium source electrode can include a complex containing any of the aforementioned materials. In some embodiments, the lithium source electrode can be coated with any of the aforementioned materials.

[0020] In some embodiments, the lithium source can include defect, scrap, or end-of-life lithium-ion batteries. In some embodiments, the lithium source can include used or end-of-life electrode sheets. In some embodiments, the lithium source can include electrode scrap materials. In some embodiments, the lithium source can include used, defect, scrap, or end-of-life lithium-ion batteries. In some embodiments, the electrode scrap materials can include cathode, anode, and/or battery work-in-process slurries. In some embodiments, the electrode scrap materials can include other battery components or work-in-process materials, or a mixture thereof. In some embodiments, the lithium source can include crushed and/or shredded end-of-life battery materials. In some embodiments, the lithium source can include a mixture of the aforementioned electrode forms.

[0021] Step **12** includes oxidation and extraction of the lithium source. This can include oxidizing (via an oxidation agent) the transition metals in the lithium source via one or more oxidation reagents. The lithium is then extracted from the suspension as an extracted lithium solution. The extracted lithium solution can be separated from the residual solids of the lithium source. In some embodiments, the oxidation agent can include $Na_2S_2O_8$. $Na_2S_2O_8$ has a standard redox potential of 1.96 V. The high oxidation voltage of $Na_2S_2O_8$ can result in less extraction time compared to other oxidizers. In some embodiments, the oxidation agent can include $K_2S_2O_8$, $(NH_4)_2S_2O_8$, or any combination thereof. In some embodiments, the oxidation agent can include H_2O_2. Hydrogen peroxide has a standard redox potential of 1.763 V. The use of H_2O_2 can significantly decrease the sodium/potassium concentration solution, which can introduce less sodium impurity to the produced lithium carbonate. Due to the reduced concentration of sodium salts when H_2O_2 is used as the oxidizer, the lithium can also be precipitated (i.e., in step **16** described below) with a smaller final volume (as less liquid is needed to prevent sodium compound precipitation). Thus, the Li_2CO_3 yield can be increased by reducing the amount of Li_2CO_3 dissolved in a solution with a smaller volume. In some embodiments, the oxidation agent can include O_3. Ozone (O_3) has a standard redox potential of 2.07 V. In some embodiments, the oxidation agent can include N_2O gas. N_2O gas has a standard redox potential of 1.77 V. In some embodiments, the oxidation agent used in the extraction step can include chlorine gas, which has a standard redox potential of 1.396 V. In some embodiments, the oxidation agent used in the extraction step can include any combination of the aforementioned oxidation agents. In some embodiments, multiple oxidation agents can be used in succession, or at least partially concurrently.

[0022] In some embodiments, the amount of oxidation agent used during the oxidation can be at least about 0.25, at least about 0.5, at least about 0.6, at least about 0.7, at least about 0.8, at least about 0.9, at least about 1, at least about 1.1, at least about 1.25, at least about 1.5, at least about 1.75, at least about 2, at least about 2.25, at least about 2.5, at least about 2.75, at least about 3, at least about 4, at least about 5, at least about 6, at least about 7, at least about 8, or at least about 9 times the stoichiometric amount needed to extract all lithium from the lithium source. In some embodiments, the amount of oxidation agent used during the oxidation can be no more than about 10, no more than about 9, no more than about 8, no more than about 7, no more than about 6, no more than about 5, no more than about 4, no more than about 3, no more than about 2.75, no more than about 2.5, no more than about 2.25, no more than about 2, no more than about 1.75, no more than about 1.5, no more than about 1.25, no more than about 1.1, no more than about 1, no more than about 0.9, no more than about 0.8, no more than about 0.7, no more than about 0.6, or no more than about 0.5 times the stoichiometric amount needed to extract all lithium from the lithium source. Combinations of the above-referenced stoichiometric ratios are also possible (e.g., at least about 0.25 and no more than about 10 or at least about 1 and no more than about 5), inclusive of all values and ranges therebetween. In some embodiments, the amount of oxidation agent used during the oxidation can be about 0.25, about 0.5, about 0.6, about 0.7, about 0.8, about 0.9, about 1, about 1.1, about 1.25, about 1.5, about 1.75, about 2, about 2.25, about 2.5, about 2.75, about 3, about 4, about 5, about 6, about 7, about 8, about 9, or about 10 times the stoichiometric amount needed to extract all lithium from the lithium source.

[0023] In some embodiments, the oxidation agent can oxidize water and generate oxygen gas and fee protons in the extracted lithium solution, thereby creating an acidic environment. For example, the use of $Na_2S_2O_8$ (or alternatively $K_2S_2O_8$ or $(NH_4)_2S_2O_8$) as the oxidizer generates H_2SO_4. The transition metals in the lithium source can gradually dissolve into the lithium solution under an acidic environment. A high concentration of these impurity ions leads to more precipitates during the purification at step **15** (description below), and results in a slower extraction speed (e.g., the filtration process, if used in step, is slower). Additionally, the dissolution of transition metals into the lithium solution can degrade the structure of the lithium-extracted electrode materials and can have a negative effect on their recyclability (e.g., recycling of $FePO_4$ for extraction from LFP source materials). In some embodiments, a mixture of $Na_2S_2O_8$ (or alternatively $K_2S_2O_8$ or $(NH_4)_2S_2O_8$) and H_2O_2 is used for the oxidation extraction. The ratio of $Na_2S_2O_8$ and H_2O_2 can be adjusted to serve different extraction preferences.

[0024] In some embodiments, the wt: wt ratio of $Na_2S_2O_8$ to H_2O_2 can be at least about 1:10, at least about 1:9, at least about 1:8, at least about 1:7, at least about 1:6, at least about 1:5, at least about 1:4, at least about 1:3, at least about 1:2, at least about 1:1, at least about 2:1, at least about 3:1, at least about 4:1, at least about 5:1, at least about 6:1, at least about 7:1, at least about 8:1, or at least about 9:1. In some embodiments, the wt: wt ratio of $Na_2S_2O_8$ to H_2O_2 can be no more than about 10:1, no more than about 9:1, no more than about 8:1, no more than about 7:1, no more than about 6:1, no more than about 5:1, no more than about 4:1, no more than about 3:1, no more than about 2:1, no more than about 1:1, no more than about 1:2, no more than about 1:3, no more than about 1:4, no more than about 1:5, no more than about 1:6, no more than about 1:7, no more than about 1:8, or no more than about 1:9. Combinations of the above-referenced weight ratios are also possible (e.g., at least about 1:10 and no more than about 10:1 or at least about 1:3 and no more than about 3:1), inclusive of all values and ranges therebetween. In some embodiments, the wt: wt ratio of $Na_2S_2O_8$ to H_2O_2 can be about 1:10, about 1:9, about 1:8, about 1:7, about 1:6, about 1:5, about 1:4, about 1:3, about 1:2, about 1:1, about 2:1, about 3:1, about 4:1, about 5:1, about 6:1, about 7:1, about 8:1, about 9:1, or about 10:1.

[0025] With a higher proportion of $Na_2S_2O_8$ to H_2O_2, the extraction speed can be faster, and the dissolution of transition metals decreases, due to the decreased pH. In some embodiments, the pH of the extracted lithium solution can be controlled during the oxidation extraction process to decrease the dissolution of transition medals. In some embodiments, the pH of the extracted lithium solution can be monitored intermittently or constantly during the oxidation extraction. In some embodiments, alkaline and/or acidic reagents can be added to the extracted lithium solution to keep the pH of the extracted solution neutral. In some embodiments, the pH of the extracted lithium solution can be controlled between about 2 and about 3, between about 3 and about 4, between about 4 and about 5, between about 5 and about 6, between about 6 and about 7, between about

7 and about 8, or between about 8 and about 9, inclusive of all values and ranges therebetween.

[0026] In some embodiments, acetic acid can be used to control the pH of the extracted lithium solution between about 3.8 and about 5.8. In some embodiments, a mixture of two or more substances from citric acid, monopotassium phosphate, dipotassium phosphate, boric acid, acetic acid, monosodium phosphate, and disodium phosphate, can be used as the pH buffer to control the pH of the extracted lithium solution over a wide range of about 3 to about 9 or even broader. In some embodiments, monopotassium phosphate (KH_2PO_4) and/or monosodium phosphate (NaH_2PO_4) can be used to control the pH of the extracted lithium solution between about 6.2 and about 8.2. In some embodiments, 25 parts of 0.2 mol/L K_2HPO_4 mixed with 23.6 parts of 0.1 mol/L of sodium hydroxide and diluted to 100 parts with water can be used to control the pH at about 6.8. The kinetics of the oxidation reaction for some LFP derivatives can be slower than LFP. In some embodiments, the LFP derivatives can include Mn, Ni, and Co substitutions.

[0027] After the oxidation extraction is completed at step **12** is completed, lithium-extracted materials that have had lithium removed are generated. Step **13** is optional and includes further processing the lithium-extracted materials. In some embodiments, the lithium-extracted materials can be lithium deficient, or delithiated compounds, such as $Li_{1-x}FePO_4$ (for $0<x\leq1$) or similar compounds that correspond to the lithium source materials. In some embodiments, the lithium-extracted materials can be directly recycled as precursors for the production of virgin cathode materials. In some embodiments, the lithium-extracted materials can be washed with water at least once, at least twice, at least three times, or at least four times to remove any soluble lithium and sodium salt impurities from the oxidation extraction at step **12**. In some embodiments, battery-grade $FePO_4$ precursor can be recovered from the lithium-extracted materials at step **13**. In some embodiments, this precursor can be generated by separating the $FePO_4$ from other lithium-extracted materials or impurities and heating the $FePO_4$ to remove other unwanted impurities, such as binder, carbon, or other organic compounds. In some embodiments, the lithium-extracted materials, such as $FePO_4$ or $Li_{1-x}FePO_4$ can be heated to a temperature of no more than about 1,000° C., no more than about 900° C., no more than about 800° C., no more than about 700° C., no more than about 600° C., no more than about 500° C., no more than about 400° C., no more than about 300° C., no more than about 200° C., or no more than about 100° C., in air or any other gaseous environment or mixture (e.g., N_2, Ar), inclusive of all values and ranges therebetween.

[0028] In some embodiments, the lithium extracted materials can be merged with new lithium source materials (including materials that have undergone preprocessing at step **11**) and undergone the oxidation extraction at step **12** to further extract any residual lithium in the source materials. In some embodiments, the oxidation extraction at step **12** can be performed a third time or additional times to increase the extraction efficiency of the method **10**. In some embodiments, the lithium extracted materials, after the oxidation extraction at step **12**, can be rinsed and washed with water or other solvents to collect residual lithium (which can include a lithium solution or lithium-containing compounds) left in the porous structure of the lithium-extracted electrode materials. In some embodiments, the residual lithium left in the porous structure of the lithium-extracted electrode materials can be collected by centrifugal dryers.

[0029] At step **14** the lithium extracted during step **12** is separated from residual solids or impurities of the lithium source. Step **15** includes precipitating and filtering these impurities, forming a purified lithium solution. Step **14** can increase the lithium purity of the extracted lithium solution formed at step **12**. Residual solids that can be removed during step **14** can include, but are not limited to Fe, Mn, Ni, Co, Mg, Ca, P, Al, and Cu. In some embodiments, these impurities can precipitate as a solid at a specific pH range and then be removed from the extracted lithium solution. In some embodiments, Fe impurities can be removed when the pH of the extracted lithium solution is adjusted to a low Fe solubility pH (e.g., a pH of about 8). In some embodiments, oxidizing reagents (e.g., hydrogen peroxide, oxygen, nitric acid, and sodium/potassium/ammonium persulfate) can be added to the extracted lithium solution to oxidize any Fe2+ to Fe3+ to better precipitate Fe in the extracted lithium solution. In some embodiments, Cu impurities are removed when the pH of the extracted lithium solution is adjusted to a low Cu solubility pH (e.g., a pH of about 9). In some embodiments, Co impurities are removed when the pH of the extracted lithium solution is adjusted to a low Co solubility pH (e.g., a pH of about or between 10 and 12). In some embodiments, Ni impurities are removed when the pH of the extracted lithium solution is adjusted to a low Ni solubility pH (e.g., a pH of about or above 10). In some embodiments, Zn impurities are removed when the pH of the extracted lithium solution is adjusted to a low Zn solubility pH (e.g., a pH of about 10). In some embodiments, Al impurities are removed when the pH of the extracted lithium solution is adjusted to a low Al solubility pH (e.g., a pH of about 6.5). In some embodiments, the aforementioned techniques to separate and remove the precipitation from the extracted lithium solution include, but are not limited to filtration, centrifugation, sedimentation, or decanting, or any combination thereof.

[0030] The solubility of some impurities in the extracted lithium solution can increase with increasing temperature. In some embodiments, the purification at step **15** can include increasing the temperature of the environment where the purification takes place. The solubility of some impurities can decrease with increasing temperature. In some embodiments, the purification at step **15** can include increasing the temperature of the environment where the purification takes place. In some embodiments, more than one impurity can be removed via pH adjustment. In some embodiments, Fe, Cu, Co, Mg, Mn, Ni, and/or Zn can precipitated and removed at a pH of about 9 to about 10, about 10 to about 11, about 11 to about 12, about 12 to about 13, or about 13 to about 14, inclusive of all values and ranges therebetween. In some embodiments, Fe, Cu, Co, Mg, Mn, Ni, and/or Zn can precipitated and removed at a pH of at least about 9, at least about 10, at least about 11, at least about 12, or at least about 13. In some embodiments, Fe, Cu, Co, Mg, Mn, Ni, and/or Zn can precipitated and removed at a pH of no more than about 14, no more than about 13, no more than about 12, no more than about 11, or no more than about 10. Combinations of the above-referenced pH values are also possible (e.g., at least about 9 and no more than about 13 or at least about 10 and no more than about 12), inclusive of all values and ranges therebetween. In some embodiments, Fe, Cu, Co, Mg, Mn, Ni, and/or Zn can precipitated and removed at a pH of about 9, about 10, about 11, about 12, or about 13.

[0031] In some embodiments, the purification of the extracted lithium solution at step **15** can include adding a salt to the extracted lithium solution. In some embodiments, the salt can include Na_2CO_3, K_2CO_3, sodium oxalate, potassium oxalate, oxalic acid, or a combination thereof. In some embodiments, the purification of the extracted lithium solution at step **15** can include multiple adjustments of the pH of the extracted lithium solution. In some embodiments, the purification of the extracted lithium solution at step **15** can include a first pH adjustment and a second pH adjustment.

[0032] In some embodiments, the first pH adjustment can be to a pH of at least about 5, at least about 5.5, at least about 6, at least about 6.5, at least about 7, at least about 7.5, at least about 8, or at least about 8.5. In some embodiments, the first pH adjustment can be to a pH of no more than about 9, no more than about 8.5, no more than about 8, no more than about 7.5, no more than about 7, no more than about 6.5, no more than about 5, or no more than about 5. Combinations of the above-referenced pH values are also possible (e.g., at least about 5 and no more than about 9 or at least about 6 and no more than about 8), inclusive of all values and ranges therebetween. In some embodiments, the first pH adjustment can be to a pH of about 5, about 5.5, about 6, about 6.5, about 7, about 7.5, about 8, about 8.5, or about 9.

[0033] In some embodiments, the second pH adjustment can be to a pH of at least about 8, at least about 8.5, at least about 9, at least about 9.5, at least about 10, or at least about 10.5. In some embodiments, the second pH adjustment can be to a pH of no more than about 11, no more than about 10.5, no more than about 10, no more than about 9.5, no more than about 9, or no more than about 8.5. Combinations of the above-referenced pH values are also possible (e.g., at least about 8 and no more than about 11 or at least about 8.5 and no more than about 10.5), inclusive of all values and ranges therebetween. In some embodiments, the second pH adjustment can be to a pH of about 8, about 8.5, about 9, about 9.5, about 10, about 10.5, or about 11.

[0034] In some embodiments, step **15** can include filtering the extracted lithium solution to remove precipitated impurities. In some embodiments, step **15** can include evaporating at least a portion of water present in the extracted lithium solution.

[0035] In some embodiments, the extracted lithium solution can be heated to a temperature of at least about 40° C., at least about 50° C., at least about 60° C., at least about 70° C., at least about 80° C., or at least about 90° C. for the precipitation of Fe, Cu, Co, Mg, Mn, Ni, and/or Zn. In some embodiments, the extracted lithium solution can be heated to a temperature of no more than about 100° C., no more than about 90° C., no more than about 80° C., no more than about 70° C., no more than about 60° C., no more than about 50° C., no more than about 40° C., or no more than about 30° C. Combinations of the above-referenced temperatures are also possible (e.g., at least about 40° C. and no more than about 100° C. or at least about 50° C. and no more than about 80° C., inclusive of all values and ranges therebetween. In some embodiments, the extracted lithium solution can be heated to a temperature of about 40° C., about 50° C., about 60° C., about 70° C., about 80° C., about 90° C., or about 100° C. for the precipitation of Fe, Cu, Co, Mg, Mn, Ni, and/or Zn. In some embodiments, the pH of the extracted lithium solution can be adjusted via CaO, $Ca(OH)_2$, NaOH, KOH, HNO_3, HCl, H_2SO_4, or any combination thereof.

[0036] In some embodiments, phosphorus impurities can be removed from the extracted lithium solution by adding Ca salts to the solution. In some embodiments, phosphorus can precipitate as calcium phosphate (e.g., $Ca_3(PO_4)_2$), which is largely insoluble in water. The phosphorus can then be removed from the extracted lithium solution via filtration or other similar solids removal methods described herein. In some embodiments, $CaCl_2$) and/or $CaSO_4$ can be added to the extracted lithium solution added to precipitate phosphate (PO_4^{3-}). In some embodiments, the extracted lithium solution can be heated to a temperature of at least about 70° C., at least about 75° C., at least about 80° C., at least about 85° C., at least about 90° C., or at least about 95° C. when the phosphate is precipitated. In some embodiments, the extracted lithium solution can be heated to a temperature of no more than about 100° C., no more than about 95° C., no more than about 90° C., no more than about 85° C., no more than about 80° C., or no more than about 75° C. when the phosphate is precipitated. Combinations of the above-referenced temperatures are also possible (e.g., at least about 70° C. and no more than about 100° C. or at least about 75° C. and no more than about 95° C.), inclusive of all values and ranges therebetween. In some embodiments, the extracted lithium solution can be heated to a temperature of about 70° C., about 75° C., about 80° C., about 85° C., about 90° C., about 95° C., or about 100° C. when the phosphate is precipitated.

[0037] In some embodiments, the Ca salts can be added during step **12**. The phosphorus extracted during step **12** forms phosphate precipitates with the Ca salts. In embodiments, the phosphate precipitates are separated and removed via filtration, centrifugation, sedimentation and decanting, or a combination thereof.

[0038] In some embodiments, the phosphate precipitation can occur over a time period of at least about 1 hour, at least about 2 hours, at least about 3 hours, at least about 4 hours, at least about 5 hours, at least about 6 hours, at least about 7 hours, at least about 8 hours, at least about 9 hours, or at least about 10 hours. In some embodiments, the phosphate precipitation can be conducted at a pH of about 8, about 9, about 10, about 11, about 12, about 13, or about 14, inclusive of all values and ranges therebetween. In some embodiments, the phosphate precipitation can be removed from the extracted lithium solution via filtration, centrifugation, or sedimentation and decanting, or a combination thereof. In some embodiments, $CaCO_3$ can be added as a pH buffer during step **12**. The dissolved calcium ions in the extracted lithium solution can precipitate the phosphate when the aforementioned pH, temperature, and/or time is reached.

[0039] In some embodiments, calcium can be removed from the extracted lithium solution by adding carbonate to the extracted lithium solution. In some embodiments, Na_2CO_3 and/or K_2CO_3 can be added to the extracted lithium solution to precipitate the calcium. In some embodiments, the amount of carbonate added is at least about 1.1, at least about 1.2, at least about 1.3, at least about 1.4, at least about 1.5, at least about 1.6, at least about 1.7, at least about 1.8, at least about 1.9, at least about 2.0, at least about 2.1, at least about 2.2, at least about 2.3, at least about 2.4, at least about 2.5, at least about 2.6, at least about 2.7, at least about 2.8, at least about 2.9, or at least about 3.0 times of the stoichiometric amount of calcium to ensure the efficient and/or effective removal of the calcium ions. In some embodiments, the excess carbonate does not react to form lithium carbonate that is beyond the solubility of the extracted lithium solution. In some embodiments, calcium can be removed by adding a hydroxide to the extracted lithium solution. Calcium precipitates as calcium hydroxide and can then be removed from the extracted lithium solution. In some embodiments, NaOH and/or KOH can be added to the extracted lithium solution to precipitate the calcium. In some embodiments, the pH of the extracted lithium solution can be maintained at a level of at least about 9, at least about 10, at least about 11, at least about 12, at least about 13, or at least about 14 to efficiently remove the calcium ions.

[0040] In some embodiments, calcium can be removed from the extracted lithium solution by adding an oxalate source to the extracted lithium solution. In some embodiments, the oxalate source can include sodium oxalate, potassium oxalate, oxalic acid, or a combination thereof. In some embodiments, the amount of oxalate added is at least about 1.1, at least about 1.2, at least about 1.3, at least about 1.4, at least about 1.5, at least about 1.6, at least about 1.7, at least about 1.8, at least about 1.9, at least about 2.0, at least about 2.1, at least about 2.2, at least about 2.3, at least about 2.4, at least about 2.5, at least about 2.6, at least about 2.7, at least about 2.8, at least about 2.9, or at least about 3.0 times of the stoichiometric amount of calcium to ensure the efficient and/or effective removal of the calcium ions. In some embodiments, the pH of the extracted lithium solution can be maintained at a level of at least about 9, at least about 10, at least about 11, at least about 12, at least about 13, or at least about 14 to efficiently remove the calcium ions.

[0041] In some embodiments, calcium and/or magnesium ions in the extracted lithium solution are removed by running the extracted lithium solution through ion exchange resins. In some embodiments, the ion exchange resins have iminodiacetic functional groups. In some embodiments, the ion exchange resins have aminophosphonic functional groups. In some embodiments, the ion exchange resins have phosphonic and/or sulphonic acid functional groups. In some embodiments, the ion exchange resins have amino methyl phosphonic acid functional groups. In some embodiments, more than one stage of ion exchange is performed.

[0042] In some embodiments, the extracted lithium solution can be partially evaporated to decrease the volume of the extracted lithium solution. In some embodiments, the evaporation is conducted before the Ca removal step. In some embodiments, the evaporation is conducted at the end of step **15** and before step **16**. Less solution volume results in less dissolved Li_2CO_3 and increases the yield for the Li_2CO_3. In some embodiments, the volume of the purified lithium solution can be tuned, modified, or otherwise monitored to ensure it is large enough to dissolve all lithium salts at the precipitation temperature. In some embodiments, sodium and/or potassium salts can precipitate out at the end of evaporation. In some embodiments, the precipitated sodium and/or potassium salts are removed via filtration, centrifugation, or sedimentation and decanting, or a combination thereof.

[0043] In some embodiments, the lithium extracted solution can be heated to a temperature of at least about 60° C., at least about 65° C., at least about 70° C., at least about 75° C., at least about 80° C., at least about 85° C., at least about 90° C., or at least about 95° C. for the calcium precipitation. In some embodiments, the lithium extracted solution can be heated to a temperature of no more than about 100° C., no more than about 95° C., no more than about 90° C., no more than about 85° C., no more than about 80° C., no more than about 75° C., no more than about 70° C., or no more than about 65° C. for the calcium precipitation. Combinations of the above-referenced temperature ranges are also possible (e.g., at least about 60° C. and no more than about 100° C. or at least about 70° C. and no more than about 90° C.), inclusive of all values and ranges therebetween. In some embodiments, the lithium extracted solution can be heated to a temperature of about 60° C., about 65° C., about 70° C., about 75° C., about 80° C., about 85° C., about 90° C., about 95° C., or about 100° C. for the calcium precipitation.

[0044] In some embodiments, calcium precipitation can occur over a time period of at least about 1 hour, at least about 2 hours, at least about 3 hours, at least about 4 hours, at least about 5 hours, at least about 6 hours, at least about 7 hours, at least about 8 hours, at least about 9 hours, or at least about 10 hours, inclusive of all values and ranges therebetween. In some embodiments, the lithium extracted solution can be maintained at a pH of about 7, about 8, about 9, about 10, about 11, about 12, about 13, or about 14, inclusive of all values and ranges therebetween.

[0045] At step **16**, lithium is precipitated from the purified lithium solution to generate a lithium-containing compound. In some embodiments, the lithium-containing compound can include Li_2CO_3. The precipitated lithium-containing compound, (for example in the form of precipitated Li_2CO_3), is then separated and removed from the purified lithium solution. In some embodiments, Na_2CO_3 and/or K_2CO_3 can be added to the purified lithium solution to precipitate the lithium-containing compound. In some embodiments, the purified lithium solution with the Na_2CO_3 and/or K_2CO_3 solution can be filtered to remove any insoluble impurities before adding the Na_2CO_3 and/or K_2CO_3 solution to the purified lithium solution. As the temperature of the purified lithium solution increases, the solubility of Na and/or K salts increases while the solubility of Li_2CO_3 decreases.

[0046] In some embodiments, the precipitation of lithium carbonate is conducted at a temperature of at least about 70° C., at least about 75° C., at least about 80° C., at least about 85° C., at least about 90° C., or at least about 95° C. In some embodiments, the precipitation of lithium carbonate is conducted at a temperature of no more than about 95° C., no more than about 90° C., no more than about 85° C., no more than about 80° C., or no more than about 75° C. Combinations of the above-referenced temperature ranges are also possible (e.g., at least about 70° C. and no more than about 100° C. or at least about 75° C. and no more than about 90° C.), inclusive of all values and ranges therebetween. In some embodiments, the precipitation of lithium carbonate is conducted at a temperature of about 70° C., about 75° C., about 80° C., about 85° C., about 90° C., about 95° C., or about 100° C.

[0047] In some embodiments, the purified lithium solution can be partially evaporated to decrease the volume of the purified lithium solution before or during the precipitation of the lithium-containing compound. Less solution volume results in less dissolved Li_2CO_3 and increases the yield for the Li_2CO_3. In some embodiments, the volume of the purified lithium solution can be tuned, modified, or otherwise monitored to ensure it is large enough to dissolve all Na and/or K salts at the precipitation temperature, as evaporation of too much purified lithium solution can precipitate the Na and/or K salts together with the lithium carbonate. In some embodiments, carbon dioxide (CO_2) gas can be pumped into a purified lithium solution to precipitate the dissolved lithium-containing compound. CO_2 can react with hydroxides in the purified lithium solution to form carbonates, and then Li_2CO_3 is formed, which can then precipitate from the purified lithium solution. In some embodiments, the Li_2CO_3 precipitate can be separated and collected via filtration, centrifugation, sedimentation, decanting, or a combination thereof.

[0048] In some embodiments, the particle size of the precipitated lithium carbonate is controlled by tuning the conditions during the precipitating step. In some embodiments, the conditions include the precipitation temperature, the stirring speed of the solution, the way of either adding lithium solution to Na_2CO_3 and/or K_2CO_3 solution or adding Na_2CO_3 and/or K_2CO_3 solution to lithium solution, solution mixing speed. In some embodiments, a higher stirring speed precipitates smaller lithium carbonate particles. Sodium and/or potassium impurities can be encapsulated by the lithium carbonate particles. A smaller lithium carbonate particle would decrease the Na and/or K impurities. In some embodiments, the Na_2CO_3 and/or K_2CO_3 solution goes through purification steps to remove impurities. In some embodiments, the impurities include Ca and/or Mg. In some embodiments, the purification steps include the addition of oxalate and/or running the purified lithium solution through an ion-exchange resin.

[0049] In some embodiments, the purity of the lithium-containing compound precipitated from the purified lithium solution can be increased via further processing. In some embodiments, Na and/or K are the main impurities in the collected lithium-containing compound (e.g., Li_2CO_3). In some embodiments, the lithium-containing compound can be washed with water or any other suitable solvent to dissolve any residual Na and/or K salts in the lithium-containing compound. In some embodiments, the lithium-containing compound can undergo a wash process via water or any other suitable solvent.

[0050] In some embodiments, the water wash process can be carried out at an elevated temperature of at least about 70° C., at least about 75° C., at least about 80° C., at least about 85° C., at least about 90° C., or at least about 95° C. In some embodiments, the water wash process can be carried out at an elevated temperature of no more than about 95° C., no more than about 90° C., no more than about 85° C., no more than about 80° C., or no more than about 75° C. Combinations of the above-referenced temperature ranges are also possible (e.g., at least about 70° C. and no more than about 100° C. or at least about 75° C. and no more than about 90° C.), inclusive of all values and ranges therebetween. In some embodiments, the water wash process can be carried out at an elevated temperature of about 70° C., about 75° C., about 80° C., about 85° C., about 90° C., about 95° C., or about 100° C. In some embodiments, the washed lithium-containing compound can be separated and collected via filtration, centrifugation, or sedimentation and decanting, or any combination thereof. In some embodiments, the wash process can be repeated multiple times to decrease the Na and K concentration in the lithium-containing compound.

[0051] In some embodiments, the lithium carbonate can be subject to size reduction before the water wash processes or between the water wash processes. In some embodiments, the size reduction is performed via milling, crushing, shredding, or a combination thereof.

[0052] Step **17** is optional and includes isolating the lithium-containing wastewater. The lithium containing wastewater can be further processed via a wastewater recycling subsystem, described in greater detail with respect to FIG. **3**. The wastewater recycling subsystem can be used to further extract lithium from the wastewater from the precipitation and the wastewater from the purification at step **16**. Wastewater isolation and processing can increase the overall lithium extraction yield of the method **10** and reduce waste.

[0053] Step **18** is optional and includes adding one or more reagents to the lithium-containing wastewater. In some embodiments, the reagent can include a phosphate. In some embodiments, the reagent can include Na_3PO_4. In some embodiments, the reagent can include K_3PO_4. The addition of regent can include further precipitates, such as Li_3PO_4.

[0054] Step **19** is optional and includes further processing of the lithium-containing wastewater. In some embodiments, the further processing can include filtration to capture precipitates. In some embodiments, step **19** can include isolating and further processing the precipitates. In some embodiments, the precipitates can include Li_3PO_4.

[0055] In some embodiments, the method **10** can produce Li_2CO_3 with a purity of at least about 70 wt %, at least about 75 wt %, at least about 80 wt %, at least about 85 wt %, at least about 90 wt %, at least about 95 wt %, at least about 96 wt %, at least about 97 wt %, at least about 98 wt %, at least about 99 wt %, at least about 99.5 wt %, at least about 99.9 wt %, or at least about 99.99 wt %, inclusive of all values and ranges therebetween. In some embodiments, the method **10** can produce battery-grade Li_2CO_3.

[0056] In some embodiments, the produced Li_2CO_3 can include less than about 1 wt % impurities. In some embodiments, the produced Li_2CO_3 can include less than about 1 wt % sodium. In some embodiments, the produced Li_2CO_3 can include less than about 0.9 wt %, less than about 0.8 wt %, less than about 0.7 wt %, less than about 0.6 wt %, less than about 0.5 wt %, less than about 0.4 wt %, less than about 0.3 wt %, less than about 0.2 wt %, less than about 0.1 wt %, less than about 0.09 wt %, less than about 0.08 wt %, less than about 0.07 wt %, less than about 0.06 wt %, less than about 0.05 wt %, less than about 0.04 wt %, less than about 0.03 wt %, less than about 0.02 wt %, or less than about 0.01 wt % sodium, inclusive of all values and ranges therebetween.

[0057] In some embodiments, the produced Li_2CO_3 can include less than about 1 wt %, less than about 0.9 wt % calcium. In some embodiments, the produced Li_2CO_3 can include less than about 0.8 wt %, less than about 0.7 wt %, less than about 0.6 wt %, less than about 0.5 wt %, less than about 0.4 wt %, less than about 0.3 wt %, less than about 0.2 wt %, less than about 0.1 wt %, less than about 0.09 wt %, less than about 0.08 wt %, less than about 0.07 wt %, less than about 0.06 wt %, less than about 0.05 wt %, less than about 0.04 wt %, less than about 0.03 wt %, less than about 0.02 wt %, or less than about 0.01 wt % calcium, inclusive of all values and ranges therebetween.

[0058] In some embodiments, the produced Li_2CO_3 can include less than about 100 wt ppm copper In some embodiments, the produced Li_2CO_3 can include less than about 90 wt ppm, less than about 80 wt ppm, less than about 70 wt ppm, less than about 60 wt ppm, less than about 50 wt ppm, less than about 40 wt ppm, less than about 30 wt ppm, less than about 20 wt ppm, less than about 10 wt ppm, less than about 9 wt ppm, less than about 8 wt ppm, less than about 7 wt ppm, less than about 6 wt ppm, less than about 5 wt ppm, less than about 4 wt ppm, less than about 3 wt ppm, less than about 2 wt ppm, less than about 1 wt ppm, less than about 0.9 wt ppm, less than about 0.8 wt ppm, less than about 0.7 wt ppm, less than about 0.6 wt ppm, less than about 0.5 wt ppm, less than about 0.4 wt ppm, less than about 0.3 wt ppm, less than about 0.2 wt ppm, less than about 0.1 wt ppm, less than about 0.09 wt ppm, less than about 0.08 wt ppm, less than about 0.07 wt ppm, less than about 0.06 wt ppm, less than about 0.05 wt ppm, less than about 0.04 wt ppm, less than about 0.03 wt ppm, less than about 0.02 wt ppm, or less than about 0.01 wt ppm copper, inclusive of all values and ranges therebetween.

[0059] In some embodiments, the produced Li_2CO_3 can include less than about 100 wt ppm iron. In some embodiments, the produced Li_2CO_3 can include less than about 90 wt ppm, less than about 80 wt ppm, less than about 70 wt ppm, less than about 60 wt ppm, less than about 50 wt ppm, less than about 40 wt ppm, less than about 30 wt ppm, less than about 20 wt ppm, less than about 10 wt ppm, less than about 9 wt ppm, less than about 8 wt ppm, less than about 7 wt ppm, less than about 6 wt ppm, less than about 5 wt ppm, less than about 4 wt ppm, less than about 3 wt ppm, less than about 2 wt ppm, less than about 1 wt ppm, less than about 0.9 wt ppm, less than about 0.8 wt ppm, less than about 0.7 wt ppm, less than about 0.6 wt ppm, less than about 0.5 wt ppm, less than about 0.4 wt ppm, less than about 0.3 wt ppm, less than about 0.2 wt ppm, less than about 0.1 wt ppm, less than about 0.09 wt ppm, less than about 0.08 wt ppm, less than about 0.07 wt ppm, less than about 0.06 wt ppm, less than about 0.05 wt ppm, less than about 0.04 wt ppm, less than about 0.03 wt ppm, less than about 0.02 wt ppm, or less than about 0.01 wt ppm iron, inclusive of all values and ranges therebetween.

[0060] In some embodiments, the produced Li_2CO_3 can include less than about 100 wt ppm zinc. In some embodiments, the produced Li_2CO_3 can include less than about 90 wt ppm, less than about 80 wt ppm, less than about 70 wt ppm, less than about 60 wt ppm, less than about 50 wt ppm, less than about 40 wt ppm, less than about 30 wt ppm, less than about 20 wt ppm, less than about 10 wt ppm, less than about 9 wt ppm, less than about 8 wt ppm, less than about 7 wt ppm, less than about 6 wt ppm, less than about 5 wt ppm, less than about 4 wt ppm, less than about 3 wt ppm, less than about 2 wt ppm, less than about 1 wt ppm, less than about 0.9 wt ppm, less than about 0.8 wt ppm, less than about 0.7 wt ppm, less than about 0.6 wt ppm, less than about 0.5 wt ppm, less than about 0.4 wt ppm, less than about 0.3 wt ppm, less than about 0.2 wt ppm, less than about 0.1 wt ppm, less than about 0.09 wt ppm, less than about 0.08 wt ppm, less than about 0.07 wt ppm, less than about 0.06 wt ppm, less than about 0.05 wt ppm, less than about 0.04 wt ppm, less than about 0.03 wt ppm, less than about 0.02 wt ppm, or less than about 0.01 wt ppm zinc, inclusive of all values and ranges therebetween.

[0061] In some embodiments, the produced Li_2CO_3 can include less than about 100 wt ppm aluminum. In some embodiments, the produced Li_2CO_3 can include less than about 90 wt ppm, less than about 80 wt ppm, less than about 70 wt ppm, less than about 60 wt ppm, less than about 50 wt ppm, less than about 40 wt ppm, less than about 30 wt ppm, less than about 20 wt ppm, less than about 10 wt ppm, less than about 9 wt ppm, less than about 8 wt ppm, less than about 7 wt ppm, less than about 6 wt ppm, less than about 5 wt ppm, less than about 4 wt ppm, less than about 3 wt ppm, less than about 2 wt ppm, less than about 1 wt ppm, less than about 0.9 wt ppm, less than about 0.8 wt ppm, less than about 0.7 wt ppm, less than about 0.6 wt ppm, less than about 0.5 wt ppm, less than about 0.4 wt ppm, less than about 0.3 wt ppm, less than about 0.2 wt ppm, less than about 0.1 wt ppm, less than about 0.09 wt ppm, less than about 0.08 wt ppm, less than about 0.07 wt ppm, less than about 0.06 wt ppm, less than about 0.05 wt ppm, less than about 0.04 wt ppm, less than about 0.03 wt ppm, less than about 0.02 wt ppm, or less than about 0.01 wt ppm aluminum, inclusive of all values and ranges therebetween.

[0062] In some embodiments, the produced Li_2CO_3 can include less than about 100 wt ppm nickel. In some embodiments, the produced Li_2CO_3 can include less than about 90 wt ppm, less than about 80 wt ppm, less than about 70 wt ppm, less than about 60 wt ppm, less than about 50 wt ppm, less than about 40 wt ppm, less than about 30 wt ppm, less than about 20 wt ppm, less than about 10 wt ppm, less than about 9 wt ppm, less than about 8 wt ppm, less than about 7 wt ppm, less than about 6 wt ppm, less than about 5 wt ppm, less than about 4 wt ppm, less than about 3 wt ppm, less than about 2 wt ppm, less than about 1 wt ppm, less than about 0.9 wt ppm, less than about 0.8 wt ppm, less than about 0.7 wt ppm, less than about 0.6 wt ppm, less than about 0.5 wt ppm, less than about 0.4 wt ppm, less than about 0.3 wt ppm, less than about 0.2 wt ppm, less than about 0.1 wt ppm, less than about 0.09 wt ppm, less than about 0.08 wt ppm, less than about 0.07 wt ppm, less than about 0.06 wt ppm, less than about 0.05 wt ppm, less than about 0.04 wt ppm, less than about 0.03 wt ppm, less than about 0.02 wt ppm, or less than about 0.01 wt ppm nickel, inclusive of all values and ranges therebetween. In some embodiments, the method **10** can produce other lithium-containing compounds, such as LiOH, lithium phosphate (Li_3PO_4), and/or lithium chloride (LiCl).

[0063] In some embodiments, the produced Li_2CO_3 can include less than about 100 wt ppm non-lithium alkali metal. In some embodiments, the produced Li_2CO_3 can include less than about 90 wt ppm, less than about 80 wt ppm, less than about 70 wt ppm, less than about 60 wt ppm, less than about 50 wt ppm, less than about 40 wt ppm, less than about 30 wt ppm, less than about 20 wt ppm, less than about 10 wt ppm, less than about 9 wt ppm, less than about 8 wt ppm, less than about 7 wt ppm, less than about 6 wt ppm, less than about 5 wt ppm, less than about 4 wt ppm, less than about 3 wt ppm, less than about 2 wt ppm, less than about 1 wt ppm, less than about 0.9 wt ppm, less than about 0.8 wt ppm, less than about 0.7 wt ppm, less than about 0.6 wt ppm, less than about 0.5 wt ppm, less than about 0.4 wt ppm, less than about 0.3 wt ppm, less than about 0.2 wt ppm, less than about 0.1 wt ppm, less than about 0.09 wt ppm, less than about 0.08 wt ppm, less than about 0.07 wt ppm, less than about 0.06 wt ppm, less than about 0.05 wt ppm, less than about 0.04 wt ppm, less than about 0.03 wt ppm, less than about 0.02 wt ppm, or less than about 0.01 wt ppm non-lithium alkali metal, inclusive of all values and ranges therebetween.

[0064] In some embodiments, the produced Li_2CO_3 can include less than about 100 wt ppm alkaline earth metal. In some embodiments, the produced Li_2CO_3 can include less than about 90 wt ppm, less than about 80 wt ppm, less than about 70 wt ppm, less than about 60 wt ppm, less than about 50 wt ppm, less than about 40 wt ppm, less than about 30 wt ppm, less than about 20 wt ppm, less than about 10 wt ppm, less than about 9 wt ppm, less than about 8 wt ppm, less than about 7 wt ppm, less than about 6 wt ppm, less than about 5 wt ppm, less than about 4 wt ppm, less than about 3 wt ppm, less than about 2 wt ppm, less than about 1 wt ppm, less than about 0.9 wt ppm, less than about 0.8 wt ppm, less than about 0.7 wt ppm, less than about 0.6 wt ppm, less than about 0.5 wt ppm, less than about 0.4 wt ppm, less than about 0.3 wt ppm, less than about 0.2 wt ppm, less than about 0.1 wt ppm, less than about 0.09 wt ppm, less than about 0.08 wt ppm, less than about 0.07 wt ppm, less than about 0.06 wt ppm, less than about 0.05 wt ppm, less than about 0.04 wt ppm, less than about 0.03 wt ppm, less than about 0.02 wt ppm, or less than about 0.01 wt ppm alkaline earth metal, inclusive of all values and ranges therebetween.

[0065] In some embodiments, the produced Li_2CO_3 can include less than about 100 wt ppm transition metals. In some embodiments, the produced Li_2CO_3 can include less than about 90 wt ppm, less than about 80 wt ppm, less than about 70 wt ppm, less than about 60 wt ppm, less than about 50 wt ppm, less than about 40 wt ppm, less than about 30 wt ppm, less than about 20 wt ppm, less than about 10 wt ppm, less than about 9 wt ppm, less than about 8 wt ppm, less than about 7 wt ppm, less than about 6 wt ppm, less than about 5 wt ppm, less than about 4 wt ppm, less than about 3 wt ppm, less than about 2 wt ppm, less than about 1 wt ppm, less than about 0.9 wt ppm, less than about 0.8 wt ppm, less than about 0.7 wt ppm, less than about 0.6 wt ppm, less than about 0.5 wt ppm, less than about 0.4 wt ppm, less than about 0.3 wt ppm, less than about 0.2 wt ppm, less than about 0.1 wt ppm, less than about 0.09 wt ppm, less than about 0.08 wt ppm, less than about 0.07 wt ppm, less than about 0.06 wt ppm, less than about 0.05 wt ppm, less than about 0.04 wt ppm, less than about 0.03 wt ppm, less than about 0.02 wt ppm, or less than about 0.01 wt ppm transition metals, inclusive of all values and ranges therebetween.

[0066] In some embodiments, the produced Li_2CO_3 can include less than about 100 wt ppm metalloids. In some embodiments, the produced Li_2CO_3 can include less than about 90 wt ppm, less than about 80 wt ppm, less than about 70 wt ppm, less than about 60 wt ppm, less than about 50 wt ppm, less than about 40 wt ppm, less than about 30 wt ppm, less than about 20 wt ppm, less than about 10 wt ppm, less than about 9 wt ppm, less than about 8 wt ppm, less than about 7 wt ppm, less than about 6 wt ppm, less than about 5 wt ppm, less than about 4 wt ppm, less than about 3 wt ppm, less than about 2 wt ppm, less than about 1 wt ppm, less than about 0.9 wt ppm, less than about 0.8 wt ppm, less than about 0.7 wt ppm, less than about 0.6 wt ppm, less than about 0.5 wt ppm, less than about 0.4 wt ppm, less than about 0.3 wt ppm, less than about 0.2 wt ppm, less than about 0.1 wt ppm, less than about 0.09 wt ppm, less than about 0.08 wt ppm, less than about 0.07 wt ppm, less than about 0.06 wt ppm, less than about 0.05 wt ppm, less than about 0.04 wt ppm, less than about 0.03 wt ppm, less than about 0.02 wt ppm, or less than about 0.01 wt ppm metalloids, inclusive of all values and ranges therebetween.

[0067] FIG. **2** is a flow diagram of a method **110** of processing an extracted lithium solution, according to an embodiment. As shown, the method **110** includes adding $CaCl_2$) to an extracted lithium solution at step **111**, adjusting the pH of the extracted lithium solution to between about 10 and about 12 at step **112**, filtering the extracted lithium solution at step **113**, adjusting the pH of the extracted lithium solution to between about 5 and about 9 at step **114**, filtering the extracted lithium solution at step **115**, optionally adding Na_2CO_3 to the extracted lithium solution at step **116**, optionally adjusting the pH of the extracted lithium solution to between about 8 and about 11 at step **117**, and optionally filtering the extracted lithium solution to form a purified lithium solution at step **118**.

[0068] Step **111** includes adding $CaCl_2$) to the extracted lithium solution. In some embodiments, the extracted lithium solution can be the same or substantially similar to the extracted lithium solution formed at step **12**, as described above with respect to FIG. **1**. In some embodiments, step **111** can include adding H_2O_2 to the extracted lithium solution. In some embodiments, the $CaCl_2$) can be added in an amount to at least react stoichiometrically with all of the dissolved phosphate in the extracted lithium solution when at least about 0.1%, at least about 1%, at least about 1%, at least about 2%, at least about 3%, at least about 4%, at least about 5%, at least about 6%, at least about 7%, at least about 8%, at least about 9%, at least about 10% of phosphate from the lithium source (e.g., originating from a LFP or LFP derivative lithium source) is leached out during oxidation extraction (e.g., at step **11**).

[0069] In some embodiments, the amount of H_2O_2 added during step **111** can at least oxidize stoichiometrically all of the Fe (II) in the extracted lithium solution to form Fe (III) when at least about 0.1%, at least about 1%, at least about 1%, at least about 2%, at least about 3%, at least about 4%, at least about 5%, at least about 6%, at least about 7%, at least about 8%, at least about 9%, at least about 10% m the lithium source (e.g., originating from a LFP or LFP derivative lithium source) is leached out as Fe (II) during the oxidation extraction (e.g., at step **11**). By adding H_2O_2, Fe (II) in the solution can be oxidized to Fe (III).

[0070] Step **112** includes adjusting the pH of the extracted lithium solution to a value between about 10 and about 12. In some embodiments, step **112** includes adjusting the pH of the extracted lithium solution to a value between about 9 and about 10. In some embodiments, the pH of the extracted lithium solution can be adjusted to a value of about 9, about 9.5, about 10, about 10.5, about 11, about 11.5, or about 12, inclusive of all values and ranges therebetween. Adjustment of the pH to such a value can precipitate Fe, Cu, Co, Mg, Mn, Ni, and/or Zn ions. In some embodiments, the adjustment of the pH to a value between about 10 and about 12 can be via the addition of NaOH and/or KOH to the extracted lithium solution. In some embodiments, phosphate ions in the extracted lithium solution can be precipitated via the addition of $CaCl_2$), (e.g., in the form of $Ca_3(PO_4)$ can be precipitated via the addition of calcium ions.

[0071] Step **113** includes filtering the extracted lithium solution to remove the precipitated impurities. Step **114** includes adjusting the pH of the extracted lithium solution to between about 9. In some embodiments, the pH of the extracted lithium solution can be to about 5, about 5.5, about 6, about 6.5, about 7, about 7.5, about 8, about 8.5, or about 9, inclusive of all values and ranges therebetween. The adjustment of the pH to a value between about 5 and about 9 can aid in precipitating aluminum impurities. In some embodiments, the pH adjustment at step **114** can be via addition of an acid to the extracted lithium solution. Step **115**

includes another repetition of filtering. The filtering at step **115** can remove the aluminum impurities from the extracted lithium solution.

[0072] Step **116** is optional and includes adding Na_2CO_3 to the extracted lithium solution. The Na_2CO_3 can aid in precipitating calcium ions from the extracted lithium solution. Step **117** is optional and includes adjusting the pH of the extracted lithium solution to a value between about 8 and about 11. In some embodiments, the adjustment can be to a value of about 8, about 8.5, about 9, about 9.5, about 10, about 10.5, or about 11, inclusive of all values and ranges therebetween. In some embodiments, the pH can be adjusted to a value of at least about 8, at least about 8.5, at least about 9, at least about 9.5, at least about 10, or at least about 10.5. In some embodiments, the adjustment of the pH can be facilitated by the addition of Na_2CO_3. The adjustment of the pH can aid in precipitating the calcium ions.

[0073] Step **118** is optional and includes filtering the extracted lithium solution to form a purified lithium solution. In some embodiments, the filtering at step **113**, step **115**, and/or step **118** can include filtration, centrifugation, sedimentation, decanting, or a combination thereof. In some embodiments, the steps of the method **110** can be performed in any sequence or with any of the steps omitted. In some embodiments, additional purification steps or repeated purification steps can be added to further increase the purity of the purified lithium solution.

[0074] In some embodiments, the pH adjustment of step **114** and the filtration of step **115** can be performed before the pH adjustment step **112** and filtration step **113**. In some embodiments, step **117** can be excluded when the pH adjustment at step **114** and the filtration at step **115** are performed before the pH adjustment at step **112** and filtration at step **113**.

[0075] In some embodiments, the method **110** can include a step that partially evaporating the extracted lithium solution. The evaporation can ensure the end volume of the extracted lithium solution is at least the amount that can dissolve all lithium salts in the solution at all temperatures between room temperature and the evaporating temperature. In some embodiments, Na and/or K salts can precipitate after evaporation. In some embodiments, the evaporation step can include removing the precipitates. In some embodiments, the precipitated Na and/or K salts are removed via filtration, centrifugation, sedimentation and decanting, or a combination thereof. In some embodiments, the evaporation step can occur before step **111**. In some embodiments, the evaporation step can occur after step **111** and before step **112**. In some embodiments, the evaporation step can occur after step **113** and before step **114**. In some embodiments, the evaporation step can occur after step **115** and before step **116**. In some embodiments, the evaporation step can occur just before the ion-exchange resin treatment described below.

[0076] In some embodiments, step **116** is optional. In some embodiments, steps **112** and **113** happen after steps **114** and **115**, and steps **116**, **117**, and **118** are optional. In some embodiments, method **110** can include a step to run the extracted lithium solution through the ion-exchange resin to remove Ca and/or Mg ions in the extracted lithium solution. In some embodiments, the ion-exchange resin treatment removes Al ions in the extracted lithium solution. In some embodiments, the ion-exchange resins are in their lithium form. In some embodiments, the ion-exchange resins are in other cation forms including Na and K. In some embodiments, a partial evaporation step is performed before the ion-exchange resin treatment. In some embodiments, the ion-exchange resin treatment happens after step **112** and step **113**. In some embodiments, steps **112** and **113** happen after steps **114** and **115**, and the ion-exchange resin treatment happens after steps **112** and **113**. In some embodiments, the ion-exchange resin treatment happens after step **118**. In some embodiments, the ion-exchange resin treatment happens at the end of the method **110**.

[0077] FIG. **3** is a flow diagram of a method **210** of processing lithium-containing wastewater, according to an embodiment. As shown, the method **210** optionally includes washing Li_2CO_3 at step **211**, adding reagent to a lithium-containing wastewater at step **212**, filtering the lithium-containing wastewater to remove the precipitated Li_3PO_4 at step **213**, filtering an extracted lithium solution to remove precipitated Li_3PO_4 at step **214**, adding Li_3PO_4 to a $CaCl_2$) solution to form a LiCl solution at step **215**, filter precipitates at step **216**, optionally adding LiCl to an extracted lithium solution at step **217**, and optionally purifying Li_2CO_3 at step **218**.

[0078] Step **211** is optional and includes washing Li_2CO_3. In some embodiments, the Li_2CO_3 can be precipitated from a lithium source (e.g., the method **10**, as described above with reference to FIG. **1**). In some embodiments, the washing can be via water. The washing can improve the purity of the $LiCO_2$ and dissolve or otherwise carry away precipitates. Step **212** includes adding a reagent to the lithium-containing wastewater. In some embodiments, the lithium-containing wastewater can originate from a Li_2CO_3 production process (e.g., the method **10**, as described above with reference to FIG. **1**). In some embodiments, the lithium-containing wastewater can include precipitation wastewater (e.g., as produced in step **16**, described above with reference to FIG. **1**) and purification wastewater (e.g., as produced from the washing process that further purifies the Li_2CO_3 product. In some embodiments, the precipitation wastewater has been subject to a precipitation step but still contains dissolved lithium. In some embodiments, the dissolved lithium can be in the form of Li_2CO_3. In some embodiments, the molar ratio of Na and/or K to Li in the precipitation wastewater can be greater than about 5:1, greater than about 6:1, greater than about 7:1, greater than about 8:1, greater than about 9:1, greater than about 10:1, greater than about 15:1, greater than about 20:1, greater than about 25:1, greater than about 30:1, or greater than about 35:1. In some embodiments, the molar ratio of Na and/or K in the precipitation wastewater can be no more than about 5:1. In some embodiments, the molar ratio of Na and/or K to Li in the precipitation wastewater can be at least about 35:1.

[0079] In some embodiments, the purification wastewater also includes dissolved lithium. In some embodiments, the dissolved lithium can be in the form of Li_2CO_3. In some embodiments, the molar ratio of Na and/or K to Li in the purification wastewater can be no more than about 4:1, no more than about 3:1, no more than about 2:1, no more than about 1:1, no more than about 0.5:1, no more than about 0.1:1, or no more than about 0.1:1. In some embodiments, the molar ratio of Na and/or K to Li in the purification wastewater can be greater than about 4:1. In some embodiments, the lithium-containing wastewater can be recycled to extract Li_2CO_3. This wastewater recycling process can also reduce the amount of overall waste from lithium extraction and can increase the overall yield of lithium carbonate from lithium extraction.

[0080] In some embodiments, the reagent added at step **212** can include a phosphate. In some embodiments, the reagent added at step **212** can include Na_3PO_4, potassium phosphate (K_3PO_4), and/or ammonium phosphate ($(NH_4)_3PO_4$). In some embodiments, the Na_3PO_4 can precipitate the Li out of the lithium-containing wastewater. In some embodiments, the Na_3PO_4 cam react with the dissolved lithium and precipitate the lithium out as Li_3PO_4. In some embodiments, the Li_3PO_4 can be precipitated at a pH of greater than about 4, greater than about 5, greater than about 6, greater than about 7, greater than about 8, greater than about 9, or greater than about 10. In some embodiments, the Li_3PO_4 can be precipitated at a pH of greater than about 11. Li_3PO_4 can be precipitated at a temperature greater than about 50° C., greater than about 60° C., greater than about 70° C., greater than about 80° C., or greater than about 90° C. In some embodiments, the phosphate added to the lithium-containing wastewater can be at least about 1.0, at least about 1.5, at least about 2.0, at least about 2.5, or at least about 3.0 times the stoichiometry amount needed to precipitate all lithium from the lithium-containing wastewater. In some embodiments, the phosphates are dissolved in water and filtered to remove any insoluble impurities before adding to the lithium-containing wastewater. In some embodiments, the dissolved Li_2CO_3 in the wastewater are first converted into more soluble lithium salts by adding acids, such as HCl or H_2SO_4, prior to the adding of Na_3PO_4. In some embodiments, the dissolved Li_2CO_3 turns into LiCl by the addition of HCl. In some embodiments, the dissolved Li_2CO_3 turns into Li_2SO_4 by the addition of H_2SO_4.

[0081] Step **213** includes filtering the lithium-containing wastewater to remove precipitated Li_3PO_4. Step **214** includes filtering the extracted lithium solution to remove precipitated Li_3PO_4. In some embodiments, the extracted lithium solution can originate from an oxidation extraction process (e.g., step **12**, as described above with reference to FIG. **1**). In some embodiments, the filtering at step **213** and/or step **214** can include filtration, centrifugation, sedimentation, decanting, or any combination thereof to isolate the Li_3PO_4. In some embodiments, the collected Li_3PO_4 can be washed with water to further remove Na and/or K impurities to produce high-purity Li_3PO_4. In some embodiments, the filtration at step **213** and/or step **214** produces Li_3PO_4 with a purity of at least about 95 wt %, at least about 96 wt %, at least about 97 wt %, at least about 98 wt %, at least about 99.5 wt %, at least about 99.6 wt %, at least about 99.7 wt %, at least about 99.8 wt %, or at least about 99.9 wt %, inclusive of all values and ranges therebetween. In some embodiments, the high-purity Li_3PO_4 can be directly reused as a precursor for the production of LFP cathode materials or LFP derivative cathode materials. In some embodiments, the high-purity Li_3PO_4 can be directly used in other industries or production processes.

[0082] At step **215**, the Li_3PO_4 is added to a $CaCl_2$) solution to convert Li_3PO_4 into a soluble LiCl solution. Step **216** includes filtration of precipitates. The Li_3PO_4 reacts with $CaCl_2$) to form soluble LiCl and $Ca_3(PO_4)_2$ precipitation. In some embodiments, the $Ca_3(PO_4)_2$ precipitation can be separated and removed by removed by filtration, centrifugation, sedimentation, or decanting, or a combination thereof, resulting in a solution containing LiCl. In some embodiments, the solution is heated to a temperature of at least about 70° C., at least about 75° C., at least about 80° C., at least about 85° C., at least about 90° C., or at least about 95° C. to convert the Li_3PO_4 into soluble LiCl. In some embodiments, the solution is heated to a temperature of no more than about 100° C., no more than about 95° C., no more than about 90° C., no more than about 85° C., no more than about 80° C., or no more than about 75° C. to convert the Li_3PO_4 into soluble LiCl. Combinations of the above-referenced temperatures are also possible (e.g., at least about 70° C. and no more than about 100° C. or at least about 75° C. and no more than about 95° C.), inclusive of all values and ranges therebetween. In some embodiments, the solution is heated to a temperature of about 70° C., about 75° C., about 80° C., about 85° C., about 90° C., about 95° C., or about 100° C. to convert the Li_3PO_4 into soluble LiCl.

[0083] In some embodiments, the formation of the soluble LiCl can occur over a time period of at least about 1 hour, at least about 2 hours, at least about 3 hours, at least about 4 hours, at least about 5 hours, at least about 6 hours, at least about 7 hours, at least about 8 hours, at least about 9 hours, or at least about 10 hours, inclusive of all values and ranges therebetween. In some embodiments, the formation of soluble LiCl can occur at a pH of about 7, about 8, about 9, about 10, about 11, about 12, about 13, or about 14, inclusive of all values and ranges therebetween. In some embodiments, no detectable Na and/or K can be present in the LiCl solution. In some embodiments, the molar ratio of Na and/or K to Li in the LiCl solution can be less than about 1.5:1, less than about 1:1, less than about 0.5:1, less than about 0.4:1, less than about 0.3:1, less than about 0.2:1, less than about 0.1:1, or less than about 0.01:1.

[0084] In some embodiments, the LiCl solution can be subject to a calcium impurity removal step to remove residual calcium in the LiCl solution. In some embodiments, calcium can be removed by reacting and precipitating the calcium in solution. In some embodiments, the calcium is precipitated by adding at least one compound containing carbonate to the LiCl solution. Calcium precipitates as $CaCO_3$ and is then removed from the LiCl solution. In some embodiments, Na_2CO_3 and/or K_2CO_3 can be added to the LiCl solution to precipitate the calcium. In some embodiments, the amount of carbonate added to the LiCl solution can be in excess of the stoichiometric amount of calcium to ensure the efficient and/or effective removal of the calcium ions. In some embodiments, the excess carbonate does not react to form Li_2CO_3 that is beyond the solubility of the LiCl solution.

[0085] In some embodiments, calcium can be removed from the LiCl solution by adding hydroxide to the LiCl solution. Calcium can then precipitate as $Ca(OH)_2$ and is then removed form the LiCl solution. In some embodiments, NaOH and/or KOH are added to the LiCl solution to precipitate the calcium. In some embodiments, the pH of the LiCl solution can be maintained at a level of at least about 10, at least about 10.5, at least about 11, at least about 11.5, at least about 12, at least about 12.5, at least about 13, at least about 13.5, or at least about 14 to efficiently remove the calcium ions from the LiCl solution. The LiCl solution can then undergo precipitation and purification to produce a high purity Li_2CO_3. In some embodiments, the low Na and/or K concentration of the LiCl solution can allow for more evaporation or removal of water in solution during precipitation of the lithium and can result in a higher yield due to the lower amount of Li_2CO2 dissolution.

[0086] At step **217**, the LiCl solution is optionally combined with an extracted lithium solution (e.g., the extracted lithium solution formed at step **11**). The combined solution containing the LiCl solution and the extracted lithium solution can be further purified at step **218**. The impurity removal can include lithium precipitation and purification to produce lithium carbonate. In some embodiments, the lithium-containing wastewater produced from the combination of the LiCl solution and the extracted lithium solution can be recycled and further processed (e.g., as described above with reference

[0087] In some embodiments, the LiCl solution can be combined with the lithium-containing wastewater prior to the addition of the reagent to the lithium-containing wastewater at step **212**. In some embodiments, the LiCl can enhance the purification process of the Li_2CO_3. In some embodiments, remaining lithium-containing wastewater can be processed multiple times via the method **210**. This can allow more lithium to be extracted from the lithium-containing wastewater. In some embodiments, the purification wastewater described above with respect to step **212** can be recycled separately by evaporating the purification wastewater to precipitate out dissolved Li_2CO_3. In some embodiments, the final volume evaporated can be enough to dissolve all of the sodium salts in the purification wastewater.

[0088] In some embodiments, the method **210** can produce Li_2CO_3 with an overall purity of at least about 70 wt %, at least about 75 wt %, at least about 80 wt %, at least about 85 wt %, at least about 90 wt %, at least about 92 wt %, at least about 95 wt %, at least about 96 wt %, at least about 97 wt %, at least about 98 wt %, or at least about 99 wt %. In some embodiments, the method **210** can produce battery grade Li_2CO_3 with an overall purity of at least about 99.5 wt %. In some embodiments, the method **210** can produce enhanced battery grade Li_2CO_3 with an overall purity of at least about 99.9 wt %. In some embodiments, the method **210** can produce superior battery grade Li_2CO_3 with an overall purity of at least about 99.99 wt %.

[0089] In some embodiments, the produced Li_2CO_3 can include no more than about 1 wt %, no more than about 0.9 wt %, no more than about 0.8 wt %, no more than about 0.7 wt %, no more than about 0.6 wt %, no more than about 0.5 wt %, no more than about 0.4 wt %, no more than about 0.3 wt %, no more than about 0.2 wt %, no more than about 0.1 wt %, no more than about 0.09 wt %, no more than about 0.08 wt %, no more than about 0.07 wt %, no more than about 0.06 wt %, no more than about 0.05 wt %, no more than about 0.04 wt %, no more than about 0.03 wt %, no more than about 0.02 wt %, or no more than about 0.01 wt % sodium.

[0090] In some embodiments, the produced Li_2CO_3 can include no more than about 1 wt %, no more than about 0.9 wt %, no more than about 0.8 wt %, no more than about 0.7 wt %, no more than about 0.6 wt %, no more than about 0.5 wt %, no more than about 0.4 wt %, no more than about 0.3 wt %, no more than about 0.2 wt %, no more than about 0.1 wt %, no more than about 0.09 wt %, no more than about 0.08 wt %, no more than about 0.07 wt %, no more than about 0.06 wt %, no more than about 0.05 wt %, no more than about 0.04 wt %, no more than about 0.03 wt %, no more than about 0.02 wt %, or no more than about 0.01 wt % calcium.

[0091] In some embodiments, the produced Li_2CO_3 can include no more than about 100 wt ppm, no more than about 90 wt ppm, no more than about 80 wt ppm, no more than about 70 wt ppm, no more than about 60 wt ppm, no more than about 50 wt ppm, no more than about 40 wt ppm, no more than about 30 wt ppm, no more than about 20 wt ppm, no more than about 10 wt ppm, no more than about 9 wt ppm, no more than about 8 wt ppm, no more than about 7 wt ppm, no more than about 6 wt ppm, no more than about 5 wt ppm, no more than about 4 wt ppm, no more than about 3 wt ppm, no more than about 2 wt ppm, no more than about 1 wt ppm, no more than about 0.9 wt ppm, no more than about 0.8 wt ppm, no more than about 0.7 wt ppm, no more than about 0.6 wt ppm, no more than about 0.5 wt ppm, no more than about 0.4 wt ppm, no more than about 0.3 wt ppm, no more than about 0.2 wt ppm, no more than about 0.1 wt ppm no more than about 0.09 wt ppm, no more than about 0.08 wt ppm, no more than about 0.07 wt ppm, no more than about 0.06 wt ppm, no more than about 0.05 wt ppm, no more than about 0.04 wt ppm, no more than about 0.03 wt ppm, no more than about 0.02 wt ppm, or no more than about 0.01 wt ppm copper.

[0092] In some embodiments, the produced Li_2CO_3 can include no more than about 100 wt ppm, no more than about 90 wt ppm, no more than about 80 wt ppm, no more than about 70 wt ppm, no more than about 60 wt ppm, no more than about 50 wt ppm, no more than about 40 wt ppm, no more than about 30 wt ppm, no more than about 25 wt ppm, no more than about 20 wt ppm, no more than about 10 wt ppm, no more than about 9 wt ppm, no more than about 8 wt ppm, no more than about 7 wt ppm, no more than about 6 wt ppm, no more than about 5 wt ppm, no more than about 4 wt ppm, no more than about 3 wt ppm, no more than about 2 wt ppm, no more than about 1 wt ppm, no more than about 0.9 wt ppm, no more than about 0.8 wt ppm, no more than about 0.7 wt ppm, no more than about 0.6 wt ppm, no more than about 0.5 wt ppm, no more than about 0.4 wt ppm, no more than about 0.3 wt ppm, no more than about 0.2 wt ppm, no more than about 0.1 wt ppm no more than about 0.09 wt ppm, no more than about 0.08 wt ppm, no more than about 0.07 wt ppm, no more than about 0.06 wt ppm, no more than about 0.05 wt ppm, no more than about 0.04 wt ppm, no more than about 0.03 wt ppm, no more than about 0.02 wt ppm, or no more than about 0.01 wt ppm iron.

[0093] In some embodiments, the produced Li_2CO_3 can include no more than about 100 wt ppm, no more than about 90 wt ppm, no more than about 80 wt ppm, no more than about 70 wt ppm, no more than about 60 wt ppm, no more than about 50 wt ppm, no more than about 40 wt ppm, no more than about 30 wt ppm, no more than about 25 wt ppm, no more than about 20 wt ppm, no more than about 10 wt ppm, no more than about 9 wt ppm, no more than about 8 wt ppm, no more than about 7 wt ppm, no more than about 6 wt ppm, no more than about 5 wt ppm, no more than about 4 wt ppm, no more than about 3 wt ppm, no more than about 2 wt ppm, no more than about 1 wt ppm, no more than about 0.9 wt ppm, no more than about 0.8 wt ppm, no more than about 0.7 wt ppm, no more than about 0.6 wt ppm, no more than about 0.5 wt ppm, no more than about 0.4 wt ppm, no more than about 0.3 wt ppm, no more than about 0.2 wt ppm, no more than about 0.1 wt ppm no more than about 0.09 wt ppm, no more than about 0.08 wt ppm, no more than about 0.07 wt ppm, no more than about 0.06 wt ppm, no more than about 0.05 wt ppm, no more than about 0.04 wt ppm, no more than about 0.03 wt ppm, no more than about 0.02 wt ppm, or no more than about 0.01 wt ppm zinc.

[0094] In some embodiments, the produced Li_2CO_3 can include no more than about 100 wt ppm, no more than about 90 wt ppm, no more than about 80 wt ppm, no more than about 70 wt ppm, no more than about 60 wt ppm, no more than about 50 wt ppm, no more than about 40 wt ppm, no more than about 30 wt ppm, no more than about 25 wt ppm, no more than about 20 wt ppm, no more than about 10 wt ppm, no more than about 9 wt ppm, no more than about 8 wt ppm, no more than about 7 wt ppm, no more than about 6 wt ppm, no more than about 5 wt ppm, no more than about 4 wt ppm, no more than about 3 wt ppm, no more than about 2 wt ppm, no more than about 1 wt ppm, no more than about 0.9 wt ppm, no more than about 0.8 wt ppm, no more than about 0.7 wt ppm, no more than about 0.6 wt ppm, no more than about 0.5 wt ppm, no more than about 0.4 wt ppm, no more than about 0.3 wt ppm, no more than about 0.2 wt ppm, no more than about 0.1 wt ppm no more than about 0.09 wt ppm, no more than about 0.08 wt ppm, no more than about 0.07 wt ppm, no more than about 0.06 wt ppm, no more than about 0.05 wt ppm, no more than about 0.04 wt ppm, no more than about 0.03 wt ppm, no more than about 0.02 wt ppm, or no more than about 0.01 wt ppm aluminum.

[0095] In some embodiments, the produced Li_2CO_3 can include no more than about 100 wt ppm, no more than about 90 wt ppm, no more than about 80 wt ppm, no more than about 70 wt ppm, no more than about 60 wt ppm, no more than about 50 wt ppm, no more than about 40 wt ppm, no more than about 30 wt ppm, no more than about 25 wt ppm, no more than about 20 wt ppm, no more than about 10 wt ppm, no more than about 9 wt ppm, no more than about 8 wt ppm, no more than about 7 wt ppm, no more than about 6 wt ppm, no more than about 5 wt ppm, no more than about 4 wt ppm, no more than about 3 wt ppm, no more than about 2 wt ppm, no more than about 1 wt ppm, no more than about 0.9 wt ppm, no more than about 0.8 wt ppm, no more than about 0.7 wt ppm, no more than about 0.6 wt ppm, no more than about 0.5 wt ppm, no more than about 0.4 wt ppm, no more than about 0.3 wt ppm, no more than about 0.2 wt ppm, no more than about 0.1 wt ppm no more than about 0.09 wt ppm, no more than about 0.08 wt ppm, no more than about 0.07 wt ppm, no more than about 0.06 wt ppm, no more than about 0.05 wt ppm, no more than about 0.04 wt ppm, no more than about 0.03 wt ppm, no more than about 0.02 wt ppm, or no more than about 0.01 wt ppm nickel.

EXAMPLES

[0096] Example 1: 1,000 g of LFP scrap electrode (consisting of LFP cathode, binder, and carbon) was preprocessed through cutting of the LFP scrap electrode into 1.5 cm LFP scrap flakes and immersed in 2.5 L of water to form a mixture. Then 712 g of $Na_2S_2O_8$ powder was added to the mixture to perform an oxidation extraction step. The LFP scrap flakes were stirred in the solution for 40 min. The oxidation reaction was observed through the elevation of temperature of the solution to about 70° C. The extracted lithium solution was separated from the lithium-extracted LFP scrap electrode materials by filtration and showed a pH of about 3.5. Next, the extracted lithium solution underwent an impurity removal process. The pH of the extracted lithium solution was adjusted to about 12 by adding NaOH and the color of the solution turned green due to the presence of Fe (II). Yellow precipitation was observed after adding another 68 g of $Na_2S_2O_8$ to the solution to oxidize the Fe (II) to Fe (III). Clear and purified lithium solution was collected after filtration. The inductively couple plasma mass spectrometry (ICP) elemental analysis for the purified lithium solution (compared to the initial extracted Li solution prior to purification) is shown in Table 1. The comparison shows that most of the Al, P, Ca, Mn, and Zn impurities were removed.

[0097] Next, 300 g of Na_2CO_3 were fully dissolved in 1 L of water and filtered. The Na_2CO_3 solution was added to the purified lithium solution to perform lithium precipitation. The precipitation reaction was carried out at 90-95° C. for 4 h. The initial lithium carbonate collected went through a series of water washing at room temperature to purify the lithium carbonate. The ICP elemental analysis (Table 2) for the final lithium carbonate product shows that the Li_2CO_3 has a purity of at least 99.20%. Additionally, the X-ray diffraction (XRD) spectrum of the lithium carbonate product shown in FIG. **4** matches well with the standard powder XRD spectrum of lithium carbonate.

[0098] The lithium-extracted LFP scrap electrode materials after oxidation extraction went through multiple water washes to remove any residual lithium and sodium. Although a small amount of carbon and binder can still exist in the material, all XRD patterns for $FePO_4$ are detected in the lithium-extracted LFP scrap materials as shown in FIG. **5**. This result indicates that the disclosed oxidation extraction procedures can extract lithium while also retaining the olivine structure of $FePO_4$ in the lithium-extracted materials. This recovered $FePO_4$ can be further purified, used for LFP synthesis, and/or reused in another application.

[0099] Example 2: In another example, a mixture including 7.5 g of LiMn0.8Fe0.2PO4 electrode (consisting of LiMn0.8Fe0.2PO4 cathode, binder, carbon, and Al current collector), 30 g of LiFePO4 scrap electrode (consisting of LifePO4 cathode, binder, carbon, and Al current collector), and 30 g of LifePO4 black mass powder (consisting of LiFePO4 cathode, graphite anode, binder, carbon and a small amount of Al/Cu impurities) was used as a mixed lithium source. The electrodes were shredded into 1.5 cm flakes and mixed with the black mass powder in 200 ml of water. After 10 min of ultrasonication, 30 g of sodium persulfate was added. The solution was heated to about 75° C. for 3 h and filtration was used to collect clear lithium solution. Next, 2 ml of 30 wt % H_2O_2 and 1.1 g of CaCl2·2H2O were added to the solution. The pH of the solution was adjusted to 12 and the solution was heated to >80° C. for 1.5 h. A clear solution was collected after filtration. To remove the Al, the pH of the solution was then adjusted to 7 with HCl acid. The solution turned cloudy and went back to a clear solution after filtration with 1-micron filter paper. Next, the pH of the solution was adjusted to 11 and 2 g of Na2CO3 was added. Purified lithium solution was collected after 2 h of mixing and then filtration. Next, 13 g of Na2CO3 were fully dissolved in 40 ml of water and filtered. The Na2CO3 solution was added to the purified lithium solution for lithium precipitation. The precipitation reaction was carried out at 90° C. for 4 h. The initial lithium carbonate collected underwent a series of water washing at >80° C. temperature. The ICP elemental analysis for the final lithium carbonate product showed that the Li2CO3 has a purity of at least 99.20%.

[0100] Another example relates to recycling lithium from lithium-containing wastewater generated during the lithium extraction process. A solution containing approximately 7.5 g of Li2CO3 and 37.5 g of NaCl dissolved in 1,100 ml water was used to represent the composition of wastewater. Hydrochloric acid was added to the solution to convert all dissolved Li_2CO_3 into LiCl, and then the pH of the solution was adjusted to about 8. Next, 28 g of Na3PO4·12H2O was added to the solution to react with the Li and precipitate out Li_3PO_4. The solution was heated to about 80° C. for 3 h, and 7.0 g of Li_3PO_4 powder was collected. The XRD pattern of the Li_3PO_4 product shown in FIG. **6** matches well with the standard powder XRD spectrum of Li_3PO_4. Next, 5.3 g of Li_3PO_4 powder was added to 100 ml of water together with 11 g of $CaCl_2·2H_2O$. The solution was heated to 85° C. for 2 h to convert the Li_3PO_4 into LiCl solution and then filtrated. The collected LiCl solution then went through a Ca removal step. Na_2CO_3 of 4 g was added to the solution and stirred for 2 h. A filtration step was administered to remove $CaCO_3$ precipitate. Finally, 3.6 g of Na_2CO_3 was fully dissolved in 20 ml of water and filtered. The Na_2CO_3 solution was added to the purified LiCl solution for lithium precipitation. The precipitation reaction was carried out at 90° C. for 4 h. The initial lithium carbonate collected underwent a series of water washing at >80° C. temperature. The ICP elemental analysis for the final lithium carbonate product showed that the Li_2CO_3 has a purity of 99.7% (Table 3).

[0101] Another example relates to recycling lithium from LFP scrap electrode materials. Approximately 600 g of LFP scrap electrode (consisting of LFP cathode, binder, and carbon) was preprocessed through cutting of the LFP scrap electrode into 1.5 cm LFP scrap flakes and immersed in 1 L of water to form a mixture. Then, 475 g of $Na_2S_2O_8$ powder was added to the mixture to perform an oxidation extraction step. The LFP scrap flakes were stirred in the solution for 40 min. The oxidation reaction was observed through the elevation of the temperature of the solution to about 70° C. The extracted lithium solution was separated from the lithium-extracted LFP scrap electrode materials by filtration and showed a pH of about 3.5. Next, the extracted lithium solution underwent an impurity removal process. Around 15 mL of H_2O_2 and 32 g of $CaCl_2$) were added to the extracted lithium solution and stirred for 2 h. The pH of the extracted lithium solution was adjusted to about 6.5 by adding NaOH. The lithium solution was filtered to remove precipitates. The pH of the extracted lithium solution was then adjusted to about 10 and stirred for 1 h. The lithium solution was filtered to remove precipitates. Na_2CO_3 was added to the solution and stirred for 1 h. The solution was filtered to remove precipitates and then partially evaporated. The extracted lithium solution was then run through the ion-exchange resin to further remove Ca ions. Clear and purified lithium solution was collected after filtration. The ICP elemental analysis for the purified lithium solution (compared to the initial extracted Li solution prior to purification) is shown in Table 4. The comparison shows that most of the Al, P, Ca, Fe, and Zn impurities were removed.

[0102] Next, 169 g of Na_2CO_3 were fully dissolved in 1 L of water and filtered. The Na_2CO_3 solution was added to the purified lithium solution to perform lithium precipitation. The precipitation reaction was carried out at 90-95° C. for 4 h. The initial lithium carbonate collected went through a series of water washing at a temperature of 85-95° C. to purify the lithium carbonate. The lithium carbonate was then fired at 300-400° C. for 2-4 h to remove the water content. The ICP elemental analysis (Table 5) shows the impurity concentrations for the final lithium carbonate product. The Li_2CO_3 has a purity of at least 99.50%.

TABLE 1

ICP Elemental Analysis for Purified Lithium Solution

	7 Li ppb	23 Na ppb	24 Mg ppb	27 Al ppb	31 P ppb	34 S ppb	35 Cl ppb	39 K ppb	44 Ca ppb	47 Ti ppb
Initial Li solution	7,398.7	29,791.4	−0.3	512.1	577.6	125.3	−0.5	39.5	45.6	2.8
Purified Li solution	9,620.1	45,961.9	2.9	2.4	1.9	162.2	0.7	120.4	−48.3	2.3

	51 V ppb	52 Cr ppb	54 Fe ppb	55 Mn ppb	59 Co ppb	60 Ni ppb	65 Cu ppb	66 Zn ppb
Initial Li solution	−0.1	−0.1	−16.9	837.3	0.1	0.9	0.3	25.6
Purified Li solution	−0.2	−0.1	−16.9	0.1	0.1	−0.6	1.1	−3.0

TABLE 2

ICP Elemental Analysis of Lithium Carbonate Product

	Na %	Mg %	Al %	Si %	P %	K %	Ca %	Ti %	V %
Li_2CO_3 product	0.0680	0.0059	0.0010	0.0051	0.0500	0.0113	0.0314	0.0055	0.0259

	Cr %	Fe %	Mn %	Co %	Ni %	Cu %	Zn %	Li_2CO_3 %
Li_2CO_3 product	0.0013	0.0304	0.0001	0.0005	0.0013	0.0009	0.0026	>99.20

TABLE 3

ICP Elemental Analysis of Lithium Carbonate Product

	Na %	Mg %	Al %	Si %	P %	K %	Ca %	Ti %	V %
Li_2CO_3 product	0.0466	0.0109	0.0071	0.0433	0.0368	0.0065	0.0629	0.0030	0.0015

	Cr %	Fe %	Mn %	Co %	Ni %	Cu %	Zn %	Li_2CO_3 %
Li_2CO_3 product	0.0001	0.0334	0.0005	0.0003	0.0033	0.0006	0.0039	>99.20

TABLE 4

ICP Elemental Analysis for Purified Lithium Solution

	7 Li ppb	23 Na ppb	24 Mg ppb	27 Al ppb	31 P ppb	34 S ppb	35 Cl ppb	39 K ppb	44 Ca ppb	47 Ti ppb
Initial Li solution	16,975.0	72,129.4	4.8	4,607.5	4,365.0	275.9	529.4	42.6	173.7	15.2
Purified Li solution	24,322.3	99,649.1	7.0	0	1.7	327.5	603.8	546.7	26.4	5.8

	51 V ppb	52 Cr ppb	54 Fe ppb	55 Mn ppb	59 Co ppb	60 Ni ppb	65 Cu ppb	66 Zn ppb
Initial Li solution	23.5	0.3	5,952.7	6.7	0.6	2.7	3.0	1.6
Purified Li solution	3.9	0.1	0	0.1	0.1	0.1	0.4	0.9

TABLE 5

ICP Elemental Analysis of Lithium Carbonate Product

	Na %	Mg %	Al %	Si %	P %	K %	Ca %	Ti %	V %
Li_2CO_3 product	0.0117	0.0006	0.0003	0.0433	0.0020	0.0004	0.0032	0.0000	0.0000

	Cr %	Fe %	Mn %	Co %	Ni %	Cu %	Zn %	Li_2CO_3 %
Li_2CO_3 product	0.0000	0.0008	0.0000	0.0000	0.0000	0.0001	0.0002	>99.50

[0103] Various concepts may be embodied as one or more methods, of which at least one example has been provided. The acts performed as part of the method may be ordered in any suitable way. Accordingly, embodiments may be constructed in which acts are performed in an order different than illustrated, which may include performing some acts simultaneously, even though shown as sequential acts in illustrative embodiments. Put differently, it is to be understood that such features may not necessarily be limited to a particular order of execution, but rather, any number of threads, processes, services, servers, and/or the like that may execute serially, asynchronously, concurrently, in parallel, simultaneously, synchronously, and/or the like in a manner consistent with the disclosure. As such, some of these features may be mutually contradictory, in that they cannot be simultaneously present in a single embodiment. Similarly, some features are applicable to one aspect of the innovations, and inapplicable to others.

[0104] In addition, the disclosure may include other innovations not presently described. Applicant reserves all rights in such innovations, including the right to embodiment such innovations, file additional applications, continuations, continuations-in-part, divisionals, and/or the like thereof. As such, it should be understood that advantages, embodiments, examples, functional, features, logical, operational, organizational, structural, topological, and/or other aspects of the disclosure are not to be considered limitations on the disclosure as defined by the embodiments or limitations on equivalents to the embodiments. Depending on the particular desires and/or characteristics of an individual and/or enterprise user, database configuration and/or relational model, data type, data transmission and/or network framework, syntax structure, and/or the like, various embodiments of the technology disclosed herein may be implemented in a manner that enables a great deal of flexibility and customization as described herein.

[0105] All definitions, as defined and used herein, should be understood to control over dictionary definitions, definitions in documents incorporated by reference, and/or ordinary meanings of the defined terms.

[0106] As used herein, in particular embodiments, the terms "about" or "approximately" when preceding a numerical value indicates the value plus or minus a range of 10%. Where a range of values is provided, it is understood that each intervening value, to the tenth of the unit of the lower limit unless the context clearly dictates otherwise, between the upper and lower limit of that range and any other stated or intervening value in that stated range is encompassed within the disclosure. That the upper and lower limits of these smaller ranges can independently be included in the smaller ranges is also encompassed within the disclosure, subject to any specifically excluded limit in the stated range. Where the stated range includes one or both of the limits, ranges excluding either or both of those included limits are also included in the disclosure.

[0107] The phrase "and/or," as used herein in the specification and in the embodiments, should be understood to mean "either or both" of the elements so conjoined, i.e., elements that are conjunctively present in some cases and disjunctively present in other cases. Multiple elements listed with "and/or" should be construed in the same fashion, i.e., "one or more" of the elements so conjoined. Other elements may optionally be present other than the elements specifically identified by the "and/or" clause, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, a reference to "A and/or B", when used in conjunction with open-ended language such as "comprising" can refer, in one embodiment, to A only (optionally including elements other than B); in another embodiment, to B only (optionally including elements other than A); in yet another embodiment, to both A and B (optionally including other elements); etc.

[0108] As used herein in the specification and in the embodiments, "or" should be understood to have the same meaning as "and/or" as defined above. For example, when separating items in a list, "or" or "and/or" shall be interpreted as being inclusive, i.e., the inclusion of at least one, but also including more than one, of a number or list of elements, and, optionally, additional unlisted items. Only terms clearly indicated to the contrary, such as "only one of" or "exactly one of," or, when used in the embodiments, "consisting of," will refer to the inclusion of exactly one element of a number or list of elements. In general, the term "or" as used herein shall only be interpreted as indicating exclusive alternatives (i.e., "one or the other but not both") when preceded by terms of exclusivity, such as "either," "one of," "only one of," or "exactly one of." "Consisting essentially of," when used in the embodiments, shall have its ordinary meaning as used in the field of patent law.

[0109] As used herein in the specification and in the embodiments, the phrase "at least one," in reference to a list of one or more elements, should be understood to mean at least one element selected from any one or more of the elements in the list of elements, but not necessarily including at least one of each and every element specifically listed within the list of elements and not excluding any combinations of elements in the list of elements. This definition also allows that elements may optionally be present other than the elements specifically identified within the list of elements to which the phrase "at least one" refers, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, "at least one of A and B" (or, equivalently, "at least one of A or B," or, equivalently "at least one of A and/or B") can refer, in one embodiment, to at least one, optionally including more than one, A, with no B present (and optionally including elements other than B); in another embodiment, to at least one, optionally including more than one, B, with no A present (and optionally including elements other than A); in yet another embodiment, to at least one, optionally including more than one, A, and at least one, optionally including more than one, B (and optionally including other elements); etc.

[0110] In the embodiments, as well as in the specification above, all transitional phrases such as "comprising," "including," "carrying," "having," "containing," "involving," "holding," "composed of," and the like are to be understood to be open-ended, i.e., to mean including but not limited to. Only the transitional phrases "consisting of" and "consisting essentially of" shall be closed or semi-closed transitional phrases, respectively, as set forth in the United States Patent Office Manual of Patent Examining Procedures, Section 2111.03.

[0111] While specific embodiments of the present disclosure have been outlined above, many alternatives, modifications, and variations will be apparent to those skilled in the art. Accordingly, the embodiments set forth herein are intended to be illustrative, not limiting. Various changes may be made without departing from the spirit and scope of the disclosure. Where methods and steps described above indicate certain events occurring in a certain order, those of ordinary skill in the art having the benefit of this disclosure would recognize that the ordering of certain steps may be modified and such modification are in accordance with the variations of the invention. Additionally, certain of the steps may be performed concurrently in a parallel process when possible, as well as performed sequentially as described above. The embodiments have been particularly shown and described, but it will be understood that various changes in form and details may be made.

1. A method comprising:

extracting lithium from a lithium source by mixing the lithium source with a solvent including an oxidation reagent, forming an extracted lithium solution;

purifying the extracted lithium solution by removing impurities to obtain a purified lithium solution; and

generating lithium carbonate from the purified lithium solution.

2. The method of claim **1**, wherein the lithium source includes at least one of defected lithium-ion batteries, manufacturing scrap of lithium-ion batteries, end-of-life lithium-ion batteries, lithium-ion battery electrode scrap, manufacturing scrap of lithium-ion battery electrode, lithium-ion battery work-in-progress materials.

3. The method of claim **1**, further comprising preprocessing the lithium source.

4. The method of claim **3**, wherein the preprocessing includes at least one of a shredding, cutting, milling, or crushing step to downsize the lithium source.

5. The method of claim **3**, wherein the preprocessing includes a heat treatment.

6. The method of claim **5**, wherein the heat treatment occurs in a controlled gas environment.

7. The method of claim **3**, wherein the preprocessing includes at least one of an ultrasonication treatment or a vacuum treatment.

8. The method of claim **1**, wherein the oxidation reagent includes at least one of sodium persulfate ($Na_2S_2O_8$), potassium persulfate ($K_2S_2O_8$), ammonium persulfate ($(NH_4)_2S_2O_8$), hydrogen peroxide (H_2O_2), ozone (O_3), or Nitrous oxide (N_2O).

9. The method of claim **1**, wherein the extracting lithium occurs in a controlled pH environment.

10. The method of claim **9**, wherein the controlled pH environment includes at least one of citric acid, monopotassium phosphate, dipotassium phosphate, boric acid, acetic acid, monosodium phosphate, or disodium phosphate.

11. The method of claim **1**, wherein forming the extracted lithium solution comprises separating the extracted lithium solution from residual solids of the lithium source.

12. The method of claim **11**, wherein separating the extracted lithium solution from residual solids of the lithium source includes at least one of filtration, centrifugation, sedimentation, or decanting.

13. The method of claim **11**, further comprising at least one of extracting more lithium from the residual solids, mixing the residual solids with a new lithium source, or further processing the residual solids to extract other materials from the residual solids.

14. The method of claim **13**, wherein further processing the residual solids comprises extracting iron phosphate from the residual solids.

15. The method of claim **1**, wherein purifying the extracted lithium solution comprises:

converting at least one soluble impurity in the extracted lithium solution to precipitated impurities; and

adjusting a pH of the extracted lithium solution.

16. The method of claim **15**, further comprising:

removing the precipitated impurities from the extracted lithium solution.

17. The method of claim **16**, wherein removing precipitated impurities from the extracted lithium solution includes at least one of filtration, centrifugation, sedimentation, or decanting.

18. The method of claim **16**, wherein removing precipitated impurities from the extracted lithium solution takes place at a temperature between about 40 degree Celsius and about 100 degree Celsius.

19. The method of claim **15**, wherein converting the at least one soluble impurity to the precipitate comprises adding at least one of calcium chloride ($CaCl_2$)) or calcium sulfate ($CaSO_4$) into the extracted lithium solution.

20. The method of claim **19**, further comprising:

adding at least one of sodium carbonate (Na_2CO_3), potassium carbonate (K_2CO_3), sodium oxalate ($Na_2C_2O_4$), potassium oxalate ($K_2C_2O_4$), oxalic acid ($H_2C_2O_4$), sodium hydroxide (NaOH), or potassium hydroxide (KOH) into the extracted lithium solution.

21. The method of claim **19**, further comprising:

removing the precipitated impurities in the form of calcium salts from the extracted lithium solution.

22. The method of claim **15**, wherein the at least one soluble impurity comprises at least one soluble transition metal impurity, the method further comprising:

oxidizing the at least one soluble transition metal impurity before converting the at least one soluble transition metal impurity to the precipitated impurities.

23. The method of claim **22**, wherein oxidizing the at least one soluble transition metal impurity comprises adding hydrogen peroxide (H_2O_2) into the extracted lithium solution.

24. The method of claim **15**, wherein adjusting pH comprises at least one pH adjustment step to adjust the pH of the extracted lithium solution to between 5 and 12.

25. The method of claim **24**, wherein adjusting the pH comprises adding at least one of sodium hydroxide (NaOH), potassium hydroxide (KOH), calcium hydroxide ($Ca(OH)_2$), calcium oxide (CaO), nitric acid (HNO_3), hydrochloric acid (HCl), or sulfuric acid (H_2SO_4).

26. The method of claim **24**, further comprising:

removing the precipitated impurities from the extracted lithium solution.

27. The method of claim **15**, further comprising:

evaporating at least a portion of water present in the extracted lithium solution.

28. The method of claim **15**, further comprising:

running the extracted lithium solution through an ion exchange resin.

29. The method of claim **1**, wherein generating lithium carbonate from the purified lithium solution comprises:

adding a carbonate source into the purified lithium solution;

extracting lithium carbonate from the purified lithium solution to obtain an extracted lithium carbonate; and

generating a lithium-containing wastewater.

30. The method of claim **29**, wherein the carbonate source includes at least one of sodium carbonate (Na_2CO_3), potassium carbonate (K_2CO_3), or carbon dioxide (CO_2).

31. The method of claim **29**, wherein extracting lithium carbonate from the purified lithium solution includes precipitating lithium carbonate from the purified lithium solution to form a precipitated lithium carbonate.

32. The method of claim **31**, wherein extracting lithium carbonate from the purified lithium solution includes separating the precipitated lithium carbonate from the purified lithium solution via at least one of filtration, centrifugation, sedimentation, or decanting.

33. The method of claim **29**, wherein extracting lithium carbonate from the purified lithium solution takes place at a temperature between about 70 degree Celsius and about 100 degree Celsius.

34. The method of claim **29**, further comprising washing the extracted lithium carbonate in water.

35. The method of claim **29**, further comprising downsizing the extracted lithium carbonate.

36. The method of claim **34**, wherein the lithium-containing wastewater includes at least one of residual lithium-containing wastewater after extracting lithium carbonate from the purified lithium solution or residual lithium-containing wastewater after washing the extracting lithium carbonate in water.

37. The method of claim **29**, further comprising:

extracting lithium from the lithium-containing wastewater.

38. The method of claim **37**, wherein extracting lithium from the lithium-containing wastewater comprises:

converting a first lithium salt in the lithium-containing wastewater to a second lithium salt;

separating the second lithium salt from the lithium-containing wastewater; and

converting the second lithium salt to a third lithium salt solution.

39. The method of claim **38**, wherein the first lithium salt includes lithium carbonate.

40. The method of claim **38**, wherein the second lithium salt includes lithium phosphate.

41. The method of claim **38**, wherein the third lithium salt includes lithium chloride.

42. The method of claim **38**, wherein converting the first lithium salt to the second lithium salt comprises adding at least one of sodium phosphate (Na_3PO_4), potassium phosphate (K_3PO_4), or ammonium phosphate ($(NH_4)_3PO_4$).

43. The method of claim **38**, wherein separating the second lithium salt includes at least one of filtration, centrifugation, sedimentation, or decanting.

44. The method of claim **38**, wherein separating the second lithium salt takes place at a temperature between about 50 degree Celsius and about 100 degree Celsius.

45. The method of claim **38**, wherein separating the second lithium salt further comprises washing the second lithium salt in water.

46. The method of claim **38**, wherein converting the second lithium salt to the third lithium salt solution comprises adding the second lithium salt into a calcium chloride solution.

47. The method of claim **38**, further comprising purifying the third lithium salt solution by removing impurities from the third lithium salt solution.

48. The method of claim **47**, wherein purifying the third lithium salt solution takes place at a temperature between about 70 degree Celsius and about 100 degree Celsius.

49. The method of claim **47**, wherein purifying the third lithium salt solution comprises:

converting a dissolved calcium salt to a precipitated calcium salt; and

removing the precipitated calcium salt from the third lithium salt solution.

50. The method of claim **38**, further comprising:

evaporating at least a portion of water present in the third lithium salt solution.

51. The method of claim **38**, further comprising generating lithium carbonate from the third lithium salt solution, wherein generating lithium carbonate comprises:

adding a carbonate source into the third lithium solution;

extracting lithium carbonate from the third lithium solution; and

generating a lithium-containing wastewater.

52. The method of claim **38**, further comprising:

feeding the third lithium salt solution to the lithium source.

53. The method of claim **38**, further comprising:

feeding the third lithium salt solution to the extracted lithium solution.

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